

Design a containerised 3

Class energy_storage	Feasibility conditional	BoM Readiness Ready	Score 7.7
-------------------------	----------------------------	------------------------	--------------

Executive Summary

Metric	Value	Unit	Confidence	Notes
Annual MWh delivered	1277.5	MWh/year	medium	Usable energy cycled once per day, adjusted for system efficiency.
Estimated CAPEX	0	GBP	low	No CAPEX claim admitted yet; supplier-backed BoM evidence is required before costs appear.
Estimated annual OPEX	0	GBP/year	low	No OPEX claim admitted yet because CAPEX has no sourced BoM basis.

Brief Intake Gate

brief_ready_for_architecture

6/6 intake areas pass; 0 review, 0 blocked. Class energy_storage (high); requirements 3; numeric facts 3; requirement trace 3/3 covered; scratch grammar yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
brief_text	pass	1	127 non-whitespace character(s) in the source brief.	none	Brief text is present.
requirement_quantification	pass	1	3 quantified requirement(s), 3 numeric fact(s).	none	Quantified requirements are sufficient for a first architecture pass.
product_class_selection	pass	1	Selected energy_storage with high confidence.	none	Product class selection is strong enough for scratch architecture.
scratch_design_support	pass	1	energy_storage has a deep scratch architecture grammar.	none	Use scratch universal architecture flow for this brief.
requirement_trace_seed	pass	1	3 covered, 0 partial, 0 uncovered requirement trace row(s).	none	Extracted requirements have trace seeds into the architecture.
assumption_boundary	pass	1	1 brief assumption note(s); no-quantified-default path: no.	none	Assumption boundary is explicit enough for first-pass architecture.

Brief Clarification Plan

no_clarification_needed

No clarification questions are needed before scratch architecture generation.

Architecture Admission Gate

architecture_generation_admitted

6/6 admission areas pass; 0 review, 0 blocked. Can use for review: yes; can proceed to BoM: yes. Source scratch_universal_architecture; next action: none.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
brief_intake_clearance	pass	1	6/6 intake areas pass; class energy_storage (high); 3 quantified requirement(s).	none	Brief intake is strong enough to admit scratch architecture generation.
clarification_clearance	pass	1	0 clarification question(s): 0 required, 0 recommended, 0 architecture-blocking.	none	No brief clarification is needed before architecture generation.
class_grammar_clearance	pass	1	Product class energy_storage; scratch universal grammar supported: yes.	none	Route this class through the scratch universal module grammar.
scratch_stage_clearance	pass	1	Architecture stage passed; source scratch_universal_architecture; modules 11; submodules 46; components 204.	none	Architecture was produced by the scratch universal architecture stage.
architecture_readiness_boundary	pass	1	Architecture ready for BoM: yes; blocking issues 0; required links 3/3.	none	Deterministic architecture readiness checks passed; sourcing work may begin only with explicit evidence.
bom_provenance_hold	pass	1	0 admitted sourcing evidence row(s), 0 source-backed BoM claim(s), 0 provenance violation(s).	none	BoM fields remain behind the explicit sourcing-evidence boundary.

Stage Integrity Gate

stage_trace_accepted

5/5 stage-integrity areas pass; 0 review, 0 blocked. Stages 8/8; ordered: yes; architecture source scratch_universal_architecture; admitted priced lines 0; provenance violations 0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
stage_sequence	pass	1	8/8 expected compiler stages are present in order.	none	Stage sequence matches the scratch compiler contract.
stage_payloads	pass	1	8/8 stages have metrics; 8/8 have evidence or explicit limitation; 8/8 have limitations.	none	Every stage carries metrics, evidence or explicit limitation, and limitations.
scratch_architecture_source	pass	1	Product class energy_storage; architecture source scratch_universal_architecture; scratch grammar supported: yes.	none	Architecture came from the scratch universal grammar.
architecture_before_sourcing	pass	1	Architecture ready: yes; readiness stage passed; sourcing stage warning.	none	Sourcing stage is downstream of architecture readiness and has a consistent status.
sourcing_provenance_boundary	pass	1	0 admitted source evidence row(s), 0 admitted priced line(s), 0 provenance violation(s).	none	Supplier, manufacturer, MPN and cost fields stay behind admitted sourcing evidence.

Scratch Lineage Gate

scratch_lineage_clean

6/6 lineage areas pass; 0 review, 0 blocked. Architecture source scratch_universal_architecture; design refs 438; source refs 0; forbidden refs 0; forbidden stage mentions 0; chain-v2 benchmark yes (chain-v2-adapted numeric analysis only).

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_source	pass	1	Product class energy_storage; architecture source scratch_universal_architecture; scratch grammar supported: yes.	none	Design content came from the scratch universal architecture stage.
design_provenance_refs	pass	1	853 provenance ref(s); 438 design/architecture ref(s); 0 invalid kind(s); 0 forbidden ref(s).	none	All design provenance references are in the allowed scratch provenance vocabulary.
bom_candidate_provenance	pass	1	204 candidate BoM line(s); 0 admitted source evidence row(s); 0 unprovenanced supplier/manufacturer/MPN/cost claim(s).	none	Candidate BoM fields remain unpriced or explicitly source-backed.
source_evidence_boundary	pass	1	0 source provenance ref(s); 0 outside admitted evidence boundary; 0 inside architecture content.	none	Source refs appear only in admitted source evidence, source ledger entries or matching source-backed BoM lines.
chain_v2_quarantine	pass	1	0 forbidden provenance ref(s); 0 forbidden stage-trace mention(s).	none	No chain-v2/state.json/modifier lineage appears inside generated design content or stage traces.
benchmark_isolation	pass	1	Benchmark source: chain-v2-adapted numeric analysis only; chain benchmark used: yes; policy: Benchmark uses aggregate counts only. Chain-v2 component names, prose, design content, prices and part numbers are not imported into the scratch generator.	none	Chain-v2 is isolated to numeric aggregate benchmark counts; no design content import is allowed.

Architecture Freeze Gate

architecture_freeze_review_required

2/6 freeze areas pass; 4 review, 0 blocked. Modules/submodules/components 11/46/204. Structurally ready for sourcing yes; independent review accepted no. Module engineering module_engineering_review_required; interfaces interface_review_ready; requirements coverage_review_ready; next action: Treat the architecture as review-ready until module review questions and critical sourcing blocks close..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
admission_and_lineage	pass	1	Architecture admission architecture_generation_admitted; stage integrity stage_trace_accepted; scratch lineage scratch_lineage_clean.	none	Architecture was admitted from the scratch flow with clean stage and lineage evidence.
module_structure	review	0.81	Component allocation allocation_ready; module engineering module_engineering_review_required; 0/11 modules pass; 5 modules still have critical sourcing blocks.	none	Treat the architecture as review-ready until module review questions and critical sourcing blocks close.
interface_contracts	review	1	Interface verification interface_review_ready; 3/3 required interfaces have carriers; 0 accepted.	none	Collect or accept interface-review evidence before calling the architecture fully frozen.
requirement_coverage	review	1	Requirement coverage coverage_review_ready; 3/3 requirements structurally ready or accepted; accepted ratio 0.	none	Requirement coverage is structurally ready; collect accepted reviewer evidence before full freeze.
engineering_review_state	review	0.67	Pre-BoM engineering engineering_review_ready; 4/6 areas pass; calculations needing review 1; assumptions needing review 15.	none	Keep architecture labelled review-ready until calculations, assumptions and verification evidence are accepted.
sourcing_boundary	pass	1	204 candidate BoM lines; 0 admitted sourcing evidence rows; 0 provenance violation(s); 24 critical missing source claim(s).	none	Sourcing and BoM claims remain behind explicit admitted source evidence.

Architecture Freeze Closure Plan

architecture_freeze_review_required

4 closure row(s): 4 ready, 0 blocked. Queues S/E/V/R 1/1/2/0; next row freeze-close:1:module_structure. Structurally ready for sourcing yes; independent review accepted no.

#	Area	Queue	Priority	Status	Required Evidence	Acceptance Criteria	Input Artifacts
1	module_structure	sourcing_intake	major	ready	Treat the architecture as review-ready until module review questions and critical sourcing blocks close.	Module engineering verdict is module_engineering_ready. No module has blocked allocation or unresolved critical sourcing blockers. Every module review question is accepted, revised or explicitly deferred.	*.module-engineering-gate.csv; *.component-allocation-gate.csv; *.engineering-review-pack.csv; *.sourcing-intake-template.csv
2	interface_contracts	verification_intake	minor	ready	Collect or accept interface-review evidence before calling the architecture fully frozen.	Interface verification verdict is accepted_interfaces. Every required interface has endpoint carriers and accepted reviewer evidence.	*.interface-contracts.json; *.interface-verification-gate.csv; *.verification-intake-template.csv
3	requirement_coverage	verification_intake	minor	ready	Requirement coverage is structurally ready; collect accepted reviewer evidence before full freeze.	Requirement coverage verdict is accepted_evidence. Every parsed requirement has architecture links and accepted verification evidence.	*.requirement-coverage-gate.csv; *.engineering-assurance-matrix.csv; *.verification-intake-template.csv
4	engineering_review_state	engineering_review	major	ready	Keep architecture labelled review-ready until calculations, assumptions and verification evidence are accepted.	Pre-BoM engineering verdict is engineering_accepted. Calculations, assumptions and verification activities are accepted or explicitly deferred.	*.pre-bom-engineering-gate.csv; *.engineering-calculations.csv; *.engineering-assumptions.csv; *.verification-plan.csv

component_candidates_ready_for_sourcing

7/7 candidate areas pass; 0 review, 0 blocked. Candidate lines 204; critical 6/6; worklist rows 204; duplicate groups 0; provenance violations 0; ready for sourcing yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
candidate_presence	pass	1	204 candidate BoM line(s), 204 distinct component identity key(s).	none	Component candidates exist for downstream sourcing review.
candidate_identity	pass	1	204/204 candidate line(s) have non-generic identity, description and role context.	none	Candidate identities are specific enough for first-pass sourcing worklists.
quantity_basis	pass	1	204/204 candidate line(s) have positive quantity and unit.	none	Candidate quantities have positive values and explicit units.
critical_candidate_coverage	pass	1	6/6 class-critical part(s) are represented by candidate lines; 6 critical BoM line(s).	none	Class-critical parts have candidate lines before sourcing begins.
duplicate_identity_review	pass	1	0 duplicate component identity group(s), 0 involving critical lines.	none	No duplicate component identities need sourcing review.
sourcing_worklist_alignment	pass	1	204/204 unpriced candidate line(s) appear in the sourcing worklist.	none	Every unpriced candidate line is visible in the sourcing worklist.
provenance_boundary	pass	1	0 source evidence row(s), 0 source-backed claim(s), 0 provenance violation(s).	none	Candidate sourcing fields remain clean unless admitted evidence supplies them.

Sourcing Authorization Gate

sourcing_authorized

6/6 sourcing-authorization areas pass; 0 review, 0 blocked. Critical intake 6/6; full intake 204/204; admitted evidence 0; rejected evidence 0; provenance violations 0; authorized yes.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_authorization	pass	1	architecture_generation_admitted; review yes, BoM yes.	none	Architecture admission permits evidence-gated sourcing work.
component_candidate_authorization	pass	1	component_candidates_ready_for_sourcing; 204 candidate line(s), 204 sourcing-worklist row(s), 0 provenance violation(s).	none	Component candidates are concrete enough for sourcing intake.
critical_intake_scope	pass	1	6/6 critical unpriced line(s) appear in the critical sourcing intake template.	none	Critical sourcing intake scope is complete.
full_worklist_scope	pass	1	204/204 total unpriced line(s) appear in the full sourcing intake template.	none	Full sourcing intake scope matches the unpriced worklist.
evidence_admission_boundary	pass	1	0 admitted source evidence row(s), 0 rejected source evidence row(s).	none	No rejected sourcing evidence is waiting for repair.
provenance_boundary	pass	1	0 source-backed BoM claim(s), 0 provenance violation(s).	none	Supplier, manufacturer, MPN, lead-time and cost fields remain behind admitted evidence.

BoM Admission Gate

candidate_bom_authorized

Display mode candidate_only. 2/7 BoM-admission areas pass; 5 review, 0 blocked. Critical priced 0/6; source-backed claims 0; provenance violations 0. Candidate BoM yes; priced BoM no; procurement use no.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_authorization	pass	1	sourcing_authorized; authorized yes; 6/6 areas pass.	none	Sourcing is authorized, so the BoM may be shown as a candidate sourcing queue.
candidate_bom_scope	pass	1	204 candidate BoM line(s), 6 critical line(s).	none	Candidate BoM scope exists for sourcing and evidence collection.
source_admission_state	review	0	0 admitted source row(s), 0 rejected source row(s), 0/204 priced line(s).	none	No source rows are admitted; keep this as a candidate-only BoM.
critical_bom_admission	review	0	0/6 critical line(s) have admitted unit costs; 6 critical line(s) remain unpriced.	none	Start with critical sourcing intake before treating any BoM cost as admitted.
source_reference_quality	review	0	no_sourcing_evidence; 0/0 source-quality rows pass; 0 protocol fixture row(s).	none	No source evidence is present; keep the BoM candidate-only.
provenance_boundary	review	0.98	0 source-backed claim(s), 24 critical missing-source claim(s), 0 provenance violation(s).	none	Missing critical source claims are acceptable only in candidate-only mode; collect evidence before priced BoM use.
display_mode	review	0.25	Display mode will be candidate_only.	none	Show candidate BoM as a sourcing queue with supplier, MPN and cost fields empty.

Report Readiness Gate

architecture_review_ready

5/6 sections at or above 8. Mean score 7.67. Architecture ready: yes. Critical unpriced lines: 6. Verification accepted: 0/25; unaccepted: 25. Verification acceptance ratio: 0.

- bom score 1 is below 8.
- 6 critical BoM line(s) lack source-backed cost evidence.
- 5 verification activity/activities remain blocked.
- 0/25 non-sourcing verification activity/activities have accepted reviewer evidence.

Section	Score	Target	B/M/m Issues	Action
executive_summary	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
brief_requirements	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
design_modules	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
bom	1	below target	6/1/0	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates.
feasibility_notes	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.
sources_references	9	pass	0/0/0	No deterministic issue currently blocks this section from the target score; external verification evidence may still be pending.

Pre-BoM Engineering Gate

engineering_review_ready

4/6 areas pass; 2 review, 0 blocked. Modules/submodules/components: 11/46/204. Calculations within envelope 2/3; calculation blockers 0. Assumption review rows 15; architecture blockers 0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_readiness	pass	1	11 modules, 46 submodules and 204 component words pass architecture readiness.	none	Architecture is structurally ready for engineering review.
component_allocation	pass	1	46/46 submodules ready; 6/6 critical parts allocated; 0 duplicate component groups.	none	Component candidates are allocated well enough for engineering review.
interface_verification	pass	1	3/3 required interfaces have endpoint carriers; 0 accepted with evidence.	none	Required interfaces are structurally ready for reviewer evidence.
requirement_coverage	pass	1	3/3 requirements are structurally ready or accepted; accepted evidence ratio 0.	none	Requirements are structurally ready for reviewer evidence.
calculation_envelope	review	0.67	2/3 calculations are within envelope; 1 need review; 0 block.	none	Attach reviewer evidence to calculations marked needs_review.
engineering_assumptions	review	0.29	6/21 assumptions are brief-supported or model-present; 15 need review/source evidence; 0 block architecture.	none	Review assumptions and keep BoM/source-required assumptions out of trusted claims until evidence is admitted.

BoM Costing Gate

costing_not_started

1/7 costing areas pass; 0 review, 6 blocked. Critical priced lines 0/6; unpriced critical lines 6. Source-backed claims 0; provenance violations 0; sourcing evidence 0/0 production-ready, 0 protocol. Source quality: no_sourcing_evidence (0/0 pass, 0 placeholder URL, 0 quote-anchored). BoM total £0; CAPEX £0.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
sourcing_admission	blocked	0	0/204 BoM lines have admitted pricing; rejected source records 0.	energy_storage_source_cell_string_lfp_prismatic_cells: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_storage_source_dc_string_output_dc_contactor_or_breaker: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_conversion_transduction_pcs_inverter_pcs_inverter: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. control_compute_communication_bms_master_bms_master_controller: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. safety_protection_fire_suppression_fire_detection_suppression: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 1 more	Admit source-backed supplier, MPN and unit-cost evidence before costing review.
critical_line_pricing	blocked	0	0/6 critical lines have admitted unit costs.	energy_storage_source_cell_string_lfp_prismatic_cells: LFP prismatic cells energy_storage_source_dc_string_output_dc_contactor_or_breaker: DC contactor or breaker energy_conversion_transduction_pcs_inverter_pcs_inverter: PCS inverter control_compute_communication_bms_master_bms_master_controller: BMS master controller safety_protection_fire_suppression_fire_detection_suppression: Fire suppression and fire detection	Price every critical line from admitted source evidence.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				re_suppression_fire _detection_suppres sion: Fire detection/suppressi on plus 1 more	
source_backed_pro venance	blocked	0.98	0 source-backed claims; 24 critical missing-source claims; 0 provenance violations.	energy_storage_sou rce_cell_string_lfp_ prismatic_cells/sup plier: Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. energy_storage_sou rce_cell_string_lfp_ prismatic_cells/man ufacturer: Admit source-backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. energy_storage_sou rce_cell_string_lfp_ prismatic_cells/mpn : Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. energy_storage_sou rce_cell_string_lfp_ prismatic_cells/unit _cost_gbp: Admit source-backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. energy_storage_sou rce_dc_string_outp ut_dc_contactor_or _breaker/supplier: Admit source- backed supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. plus 19 more	Keep supplier, manufacturer, MPN, unit cost and lead- time fields empty until source-backed evidence is admitted.
component_identity	pass	1	0 duplicate component identity group(s); 0 include critical lines.	none	Component identities are unique enough for line- level costing.
sourcing_evidence_ authenticity	blocked	0	0/0 sourcing evidence rows are production-ready; 0 protocol fixture, 0 review/missing metadata.	none	Add sourcing evidence rows before BoM costing can start.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
source_reference_qualified	blocked	0	0/0 sourcing refs pass quality checks; 0 placeholder URL, 0 quote-anchored, 0 fresh timestamp.	none	Add source references before costing review.
cost_totals	blocked	0	BoM total 0 GBP; CAPEX 0 GBP; annual OPEX 0 GBP.	6 critical line(s) are unpriced, so totals are incomplete.	Do not use cost totals until every critical line has an admitted unit cost.

Evidence Acquisition Plan

36 missing-evidence row(s): 9 sourcing, 27 verification. Ready 33; blocked 0. Blocks BoM 9; blocks publish 36. Next row: acquire:4:assumption:metric:capex_gbp.

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
1	sourcing	blocker	open	readiness:2	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
2	verification	blocker	open	readiness:3	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
3	verification	blocker	open	readiness:4	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
4	sourcing	blocker	ready_for_intake	assumption:metric:capex_gbp	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						sourceGrade; evidence.ref; evidence.quote; retrievedAt		
5	sourcing	blocker	ready_for_intake	assumption:metric:opex_annual_gbp	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
6	sourcing	blocker	ready_for_intake	lfp_prismatic_cells	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
7	sourcing	blocker	ready_for_intake	dc_contactor_or_breaker	supplier_catalogue_or_quote	componentWorldId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidence.ref ; evidence.quote; retrievedAt		
8	sourcing	blocker	ready_for_intake	pcs_inverter	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
9	sourcing	blocker	ready_for_intake	bms_master_controller	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
10	sourcing	blocker	ready_for_intake	fire_detection_suppression	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref ;	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						evidence.quote; retrievedAt		
11	sourcing	blocker	ready_for_intake	thermal_management_loop	supplier_catalogue_or_quote	componentWordId; supplierName; manufacturer; mpn where applicable; unitCostGbp; leadTimeWeeks where stated; sourceGrade; evidence.ref; evidence.quote; retrievedAt	LLM-estimated cost; benchmark average without supplier evidence; class-pack default; test-fixture:// source	yes
12	verification	major	ready_for_intake	design_review:energy_storage_source	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
13	verification	major	ready_for_intake	calculation:energy_storage_source	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
14	verification	major	ready_for_intake	design_review:energy_conversion_transduction	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
15	verification	major	ready_for_intake	calculation:energy_conversion_transduction	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
16	verification	major	ready_for_intake	design_review:structure_containment	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
17	verification	major	ready_for_intake	calculation:structure_containment	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
18	verification	major	ready_for_intake	design_review:sensing_instrumentation	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
19	verification	major	ready_for_intake	design_review:control_compute_communication	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
20	verification	major	ready_for_intake	calculation: control_compute_communication	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
21	verification	major	ready_for_intake	design_review:safety_protection	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
22	verification	major	ready_for_intake	design_review:environmental_interface	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
23	verification	major	ready_for_intake	design_review:power_distribution	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
24	verification	major	ready_for_intake	calculation: power_distribution	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
25	verification	major	ready_for_intake	design_review:maintenance_serviceability	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
26	verification	major	ready_for_intake	calculation: maintenance_serviceability	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
27	verification	major	ready_for_intake	design_review:mass_fluid_transport_process	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
28	verification	major	ready_for_intake	design_review:hmi_ergonomics	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
29	verification	major	ready_for_intake	interface_review:energy_storage_source:energy_conversion_transduction:dc_bus	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture:// review; source-evidence claims that belong in sourcing intake	no
30	verification	major	ready_for_intake	interface_review:energy_storage_source:environ	named_engineering_review	activityId; evidenceKind; reviewerNa	anonymous reviewer; unchecked model	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
				mental_interface:thermal_loop		me; verdict; evidenceRef; evidenceNote; reviewedAt	assertion; test-fixture://review; source-evidence claims that belong in sourcing intake	
31	verification	major	ready_for_intake	interface_review:energy_storage_source:control_compute_communication:bms_network	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture://review; source-evidence claims that belong in sourcing intake	no
32	verification	major	ready_for_intake	compliance_review:IEC-62619	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture://review; source-evidence claims that belong in sourcing intake	no
33	verification	major	ready_for_intake	compliance_review:UL-9540A	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNote; reviewedAt	anonymous reviewer; unchecked model assertion; test-fixture://review; source-evidence claims that belong in sourcing intake	no
34	verification	major	ready_for_intake	compliance_review:G99	named_engineering_review	activityId; evidenceKind; reviewerName; verdict; evidenceRef; evidenceNo	anonymous reviewer; unchecked model assertion; test-fixture://review; source-	no

#	Kind	Priority	Status	Subject	Target	Required Fields	Disallowed Evidence	Blocks BoM
						te; reviewedAt	evidence claims that belong in sourcing intake	
35	verification	major	ready_for_in take	risk_review: 1	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no
36	verification	major	ready_for_in take	risk_review: 2	named_engi neering_revi ew	activityId; evidenceKin d; reviewerNa me; verdict; evidenceRef ; evidenceNo te; reviewedAt	anonymous reviewer; unchecked model assertion; test- fixture:// review; source- evidence claims that belong in sourcing intake	no

Document Trust Gate

evidence_blocked

1/7 trust areas pass; 3 review, 3 blocked. Architecture ready: yes. Claim gate: evidence_blocked. Reviewer evidence 0/25. Source-backed BoM claims 0; critical missing BoM claims 24; provenance violations 0. Evidence authenticity: no_evidence (0/0 production-ready, 0 protocol).

- bom: score 1 below 8
- architecture design has 0 blocked/source-required claim(s) and 57 review-required claim(s).
- engineering math has 2 blocked/source-required claim(s) and 4 review-required claim(s).
- bom sourcing has 30 blocked/source-required claim(s) and 198 review-required claim(s).
- compliance risk has 0 blocked/source-required claim(s) and 5 review-required claim(s).
- energy_storage_source_cell_string_lfp_prismatic_cells/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- energy_storage_source_cell_string_lfp_prismatic_cells/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- energy_storage_source_cell_string_lfp_prismatic_cells/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- energy_storage_source_cell_string_lfp_prismatic_cells/unit_cost_gbp: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- energy_storage_source_dc_string_output_dc_contactor_or_breaker/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- energy_storage_source_dc_string_output_dc_contactor_or_breaker/manufacturer: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
- energy_storage_source_dc_string_output_dc_contactor_or_breaker/mpn: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture_readiness	pass	1	11 modules, 46 submodules and 3 required links pass deterministic architecture gates.	none	No deterministic architecture blocker remains; continue with reviewer evidence before treating it as accepted.
section_scores	blocked	0.83	5/6 scored sections are at or above 8; mean 7.67.	bom: score 1 below 8	Repair below-target sections through the specific score-ledger actions.
claim_evidence	blocked	0.01	3/299 claims pass; 32 blocked/source-required, 264 review-required.	architecture design has 0 blocked/source-required claim(s) and 57 review-required claim(s). engineering math has 2 blocked/source-required claim(s) and 4 review-required claim(s). bom sourcing has 30 blocked/source-required claim(s) and 198 review-required claim(s). compliance risk has 0 blocked/source-required claim(s) and 5 review-required claim(s).	Accept, correct or reject generated module and submodule allocation claims through engineering review. Accept, correct or reject calculated metrics and engineering envelopes through reviewer evidence. Admit source-backed supplier, manufacturer, MPN, unit-cost and lead-time evidence for required BoM claims. Accept, correct or reject class-pack regulatory and risk claims through compliance/domain review.
requirement_assurance	review	0	0/3 requirements have accepted assurance; 3 need review, 0 blocked, 0 unlinked.	none	Collect reviewer evidence for requirement assurance rows before publication.
reviewer_evidence	review	0	0/25 non-sourcing verification activities accepted; 0 rejected, 0 deferred, 25 pending/blocked.	none	Collect accepted reviewer evidence for every non-sourcing verification activity.
bom_provenance	blocked	0.98	0 source-backed BoM claims, 24 critical missing-source claims, 0 provenance violations.	energy_storage_source_cell_string_lfp_prismatic_cells/supplier: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_storage_source_cell_string_lfp_prismatic_cells/manufacturer: Admit	Admit source-backed evidence for critical BoM supplier, manufacturer, MPN and cost claims.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_storage_source_cell_string_lfp_prismatic_cells/mpn : Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_storage_source_cell_string_lfp_prismatic_cells/unit_cost_gbp: Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. plus 19 more	

evidence_authenticity	review	0	0/0 evidence row(s) are production-ready; 0 protocol fixture, 0 review/missing metadata.	none	Admit source and reviewer evidence before trusting the document.
-----------------------	-------------------	---	--	------	--

Evidence Authenticity Gate

no_evidence

0/0 evidence rows are production-ready. Protocol fixtures 0; review-required 0; missing metadata 0. External URLs 0; internal references 0; local files 0; unknown references 0.

Kind	Subject	Reference Class	Status	Ref	Reason	Required Action
------	---------	-----------------	--------	-----	--------	-----------------

Evidence Replacement Plan

No evidence replacement rows remain for the current authenticity verdict: no_evidence.

Trust Repair Plan

6 package(s): 6 ready, 0 waiting. Next package: sourcing_intake. Closure rows 45; trust blockers 37. Trust verdict: evidence_blocked.

#	Package	Status	Gate Areas	Closure Rows	Trust Blockers	Required Inputs	Source Artifacts	Exit Criteria
2	Admit Source-Backed BoM Claims	ready	claim_evidence; bom_provenance	9	28	Supplier, manufacturer, MPN, unit cost, lead time, retrieved-at timestamp and source reference for critical BoM lines. Rejected or duplicate component identities resolved before pricing shared items.	*.sourcing-intake-template.csv; *.sourcing-pack.csv; *.bom-provenance-manifest.csv; *.component-identity.csv	BoM provenance has zero critical missing-source claims. Claim Evidence Gate BoM sourcing area is no longer evidence_blocked.
3	Resolve Engineering Review Questions	ready	claim_evidence; requirement_assurance	8	4	Accepted/corrected/rejected answers to review-pack questions. Calculation and assumption decisions tied to reviewer evidence references.	*.engineering-review-pack.csv; *.engineering-assurance-matrix.csv; *.engineering-assumptions.csv; *.engineering-calculations.csv	Requirement assurance rows have accepted or explicitly deferred engineering review evidence. Generated architecture and calculation claims are accepted, corrected or rejected.
4	Admit Reviewer Verification Evidence	ready	reviewer_evidence	27	0	Accepted, rejected or deferred verification evidence for every non-source verification activity.	*.verification-intake-template.csv; *.verification-plan.csv; *.verification-ledger.csv	Every non-source verification activity has accepted reviewer evidence or an explicit defer decision.
5	Validate Evidence Authenticity	ready	evidence_authenticity	0	0	Production supplier/catalogue URLs for BoM source records. Named reviewer evidence references that are not protocol fixtures.	*.evidence-authenticity-gate.csv; *.sourcing-ledger.csv; *.verification-ledger.csv; *.claim-ledger.csv	Evidence Authenticity Gate verdict is production_ready. No protocol_fixture, unknown_reference or missing_metadata evidence

#	Package	Status	Gate Areas	Closure Rows	Trust Blockers	Required Inputs	Source Artifacts	Exit Criteria
								rows remain.
6	Review Compliance And Risk Claims	ready	claim_evidence	0	4	Compliance -domain review of class-pack standards, hazards and mitigation claims.	*.claim-ledger.csv; *.engineering-review-pack.csv	Compliance and risk claims are accepted, corrected or rejected by a domain reviewer.
7	Repair Residual Section Scores	ready	section_scores	1	1	A rerun after evidence intake, plus score-ledger fixes for any section still below target.	*.score-ledger.csv; *.readiness-gate.csv; *.document-trust-gate.csv	All scored sections are at or above target and Document Trust Gate is no longer blocked by scoring.

Claim Ledger

299 claims: 3 brief-supplied, 0 reviewer-accepted, 0 source-backed, 4 calculated needing review, 260 generated needing review, 32 source-required, 0 blocked. Blocking: 32 BoM, 294 publishable.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
bom_source_field	source_required	assumption	BMS master controller / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	BMS master controller / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	BMS master controller / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	BMS master controller / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	PCS inverter / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	PCS inverter / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								before BoM can pass.
bom_source_field	source_required	assumption	PCS inverter / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	PCS inverter / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	LFP prismatic cells / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	LFP prismatic cells / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	LFP prismatic cells / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	LFP prismatic cells / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								cost evidence before BoM can pass.
bom_source_field	source_required	assumption	DC contactor or breaker / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	DC contactor or breaker / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	DC contactor or breaker / supplier	supplier is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	DC contactor or breaker / unit_cost_gbp	unit_cost_gbp is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Thermal management loop / manufacturer	manufacturer is required but not source-backed.	none	yes	yes	Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Thermal management loop / mpn	mpn is required but not source-backed.	none	yes	yes	Admit source-backed supplier/ma

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
								nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Thermal managemen t loop / supplier	supplier is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Thermal managemen t loop / unit_cost_g bp	unit_cost_g bp is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Fire detection/su ppression / manufactur er	manufactur er is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Fire detection/su ppression / mpn	mpn is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Fire detection/su ppression / supplier	supplier is required but not source- backed.	none	yes	yes	Admit source- backed supplier/ma nufacturer/ MPN/unit- cost evidence before BoM can pass.
bom_source_field	source_required	assumption	Fire detection/su	unit_cost_g bp is	none	yes	yes	Admit source-

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	source_required	generated_model	ppression / unit_cost_gbp	required but not source-backed.				backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
			BMS master controller	BMS master controller is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
			PCS inverter	PCS inverter is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
			LFP prismatic cells	LFP prismatic cells is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
			DC contactor or breaker	DC contactor or breaker is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	source_required	generated_model	Thermal management loop	Thermal management loop is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
component_candidate	source_required	generated_model	Fire detection/suppression	Fire detection/suppression is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	yes	yes	Admit source-backed supplier, manufacturer, MPN, unit cost and lead-time evidence through sourcing intake.
headline_metric	source_required	assumption	Estimated CAPEX	Estimated CAPEX = 0 GBP.	formula:capex=sum(bom.totalCostGbp); class_pack:energy_storage.class_pack	yes	yes	Admit source-backed BoM costs before treating this cost metric as a claim.
headline_metric	source_required	assumption	Estimated annual OPEX	Estimated annual OPEX = 0 GBP/year.	formula:opex=capex*0.04	yes	yes	Admit source-backed BoM costs before treating this cost metric as a claim.
component_candidate	generated_needs_review	generated_model	contactor driver output	contactor driver output is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	isolation interface	isolation interface is an architecture-derived component candidate with	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
				quantity 1 candidate.				installation instance.
component_candidate	generated_needs_review	generated_model	SOC estimation firmware	SOC estimation firmware is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	cybersecurity firewall	cybersecurity firewall is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	Modbus TCP gateway	Modbus TCP gateway is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	SCADA protocol adapter	SCADA protocol adapter is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	time synchronisation module	time synchronisation module is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	generated_needs_review	generated_model	account key switch	account key switch is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	configuration backup module	configuration backup module is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	firmware escrow tag	firmware escrow tag is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	syslog collector	syslog collector is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	VPN access appliance	VPN access appliance is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	generated_needs_review	generated_model	cellular router	cellular router is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	I/O expansion module	I/O expansion module is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	industrial controller	industrial controller is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	network switch	network switch is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	AC filter capacitor	AC filter capacitor is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	generated_needs_review	generated_model	EMI filter assembly	EMI filter assembly is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	grid filter choke	grid filter choke is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	protective earth bar	protective earth bar is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	G99 settings label	G99 settings label is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	grid protection relay	grid protection relay is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	relay test block	relay test block is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	generated_needs_review	generated_model	trip circuit monitor	trip circuit monitor is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	voltage sensing fuse set	voltage sensing fuse set is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	DC link capacitor bank	DC link capacitor bank is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	gate driver board	gate driver board is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	IGBT power module	IGBT power module is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	generated_needs_review	generated_model	inverter control board	inverter control board is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	liquid-cooled heatsink	liquid-cooled heatsink is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	anti-vibration mount	anti-vibration mount is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	cast resin transformer	cast resin transformer is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	HV winding	HV winding is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Kind	Status	Basis	Scope	Claim	Evidence Refs	Blocks BoM	Blocks Publish	Next Action
component_candidate	generated_needs_review	generated_model	LV winding	LV winding is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	temperature probe	temperature probe is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	AFE monitor IC	AFE monitor IC is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	BMS slave PCB	BMS slave PCB is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.
component_candidate	generated_needs_review	generated_model	conformal coating	conformal coating is an architecture-derived component candidate with quantity 1 candidate.	model:scratch_universal_architecture.energy_storage	no	yes	Review whether this is a unique physical item, shared component or separate installation instance.

Showing first 70 of 299 claims; export CSV for the full claim ledger.

Claim Evidence Gate

evidence_blocked

3/299 claims pass by being brief-supplied, source-backed or reviewer-accepted. 32 blocked/source-required claims, 264 review-required claims. Claim pass ratio 0.01. Blocking: 32 BoM, 294 publishable.

- architecture design has 0 blocked/source-required claim(s) and 57 review-required claim(s).
- engineering math has 2 blocked/source-required claim(s) and 4 review-required claim(s).
- bom sourcing has 30 blocked/source-required claim(s) and 198 review-required claim(s).
- compliance risk has 0 blocked/source-required claim(s) and 5 review-required claim(s).

Area	Verdict	Claims	Passed	Review Req.	Blocked	Source Req.	Blocks BoM	Blocks Publish	Pass Ratio	Required Action
brief_requirements	complete	3	3	0	0	0	0	0	1	No claim-evidence action required for this area.
architecture_design	review	57	0	57	0	0	0	57	0	Accept, correct or reject generated module and submodule allocation claims through engineering review.
engineering_math	blocked	6	0	4	2	2	2	4	0	Accept, correct or reject calculated metrics and engineering envelopes through reviewer evidence.
bom_sourcing	blocked	228	0	198	30	30	30	228	0	Admit source-backed supplier, manufacturer, MPN, unit-cost and lead-time evidence for required BoM claims.
compliance_risk	review	5	0	5	0	0	0	5	0	Accept, correct or reject class-pack regulatory and risk claims through compliance

Area	Verdict	Claims	Passed	Review Req.	Blocked	Source Req.	Blocks BoM	Blocks Publish	Pass Ratio	Required Action
										ce/domain review.

Evidence Gap Register

45 open evidence gaps: 12 blockers, 33 major, 0 minor. Closure paths: 9 sourcing intake, 27 verification intake, 8 engineering review, 0 architecture revision. Blocking rows: architecture 0, BoM 9, publishable 45.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
readiness_blocker	blocker	open	Report readiness gate	bom score 1 is below 8.	score_repair	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	6 critical BoM line(s) lack source-backed cost evidence.	sourcing_intake	bom; publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	5 verification activity/activities remain blocked.	verification_intake	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier, manufacturer, MPN and cost evidence for critical BoM lines.
readiness_blocker	blocker	open	Report readiness gate	0/25 non-sourcing verification activity/activities have accepted reviewer evidence.	verification_intake	publishable	Admit source-backed evidence for every critical BoM line through sourcing intake. Increase priced-line coverage with admissible supplier evidence, not estimates. Collect engineering-review evidence for pending verification activities. Accept reviewer evidence for every non-sourcing verification activity before treating the report as publishable. Clear blocked verification activities or explicitly defer them with reviewer evidence. Use sourcing intake to admit source-backed supplier,

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
							manufacturer, MPN and cost evidence for critical BoM lines.
calculation_review	major	needs review	BESS annual nameplate throughput	1277.5 MWh/year assumes one full equivalent cycle per day.	engineering_review	publishable	Confirm cycles per year, usable depth of discharge, round-trip efficiency and degradation assumptions.
assumption_review	major	needs review	Annual MWh delivered	Annual MWh delivered uses capacity_mwh * cycles_per_year * efficiency with medium confidence.	engineering_review	publishable	Review the formula, input values and operating envelope before treating this metric as a performance claim.
assumption_review	blocker	ready for intake	Estimated CAPEX	Estimated CAPEX uses an implicit model with low confidence.	sourcing_intake	bom; publishable	Admit source-backed BoM pricing before treating this cost metric as a claim.
assumption_review	blocker	ready for intake	Estimated annual OPEX	Estimated annual OPEX uses an implicit model with low confidence.	sourcing_intake	bom; publishable	Admit source-backed BoM pricing before treating this cost metric as a claim.
assumption_review	major	needs review	BESS C-Rate	BESS C-Rate currently evaluates as pass against 0.1C to 1.0C for typical grid-storage duration systems.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	Rated Duration	Rated Duration currently evaluates as pass against 1 to 6 hours for common containerised BESS applications.	engineering_review	publishable	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	Container-Level Energy	Container-Level Energy	engineering_review	publishable	Supply an engineering

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
			Density	Density currently evaluates as pass against 50 to 200 Wh/kg at container/system level before detailed structural mass review.			calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
assumption_review	major	needs review	IEC-62619	Industrial lithium battery safety is relevant to this product class in Global / UK-adopted.	engineering_review	publishable	Cell certificates and pack-level safety validation.
assumption_review	major	needs review	UL-9540A	Thermal runaway fire propagation test is relevant to this product class in US / investor diligence.	engineering_review	publishable	Cell, module and unit-level propagation evidence.
assumption_review	major	needs review	G99	UK grid connection requirements is relevant to this product class in UK.	engineering_review	publishable	Protection settings, relay test certificates and commissioning records.
sourcing_evidence	blocker	ready for intake	LFP prismatic cells	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	DC contactor or breaker	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
sourcing_evidence	blocker	ready for intake	PCS inverter	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	BMS master controller	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Fire detection/suppression	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
sourcing_evidence	blocker	ready for intake	Thermal management loop	Required class-critical line must be sourced before BoM can pass.	sourcing_intake	bom; publishable	Supplier name, manufacturer, MPN where applicable, unit cost, lead time and source quote/reference through sourcing intake.
verification_evidence	major	ready for intake	Energy Storage Source	Review Energy Storage Source architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Energy Storage Source	Verify Energy Storage Source	verification_intake	publishable	Collect reviewer evidence through the

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
				requirement calculations			verification intake template.
verification_evidence	major	ready for intake	Power Conversion System	Review Power Conversion System architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Power Conversion System	Verify Power Conversion System requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Container Structure And Fit-Out	Review Container Structure And Fit-Out architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Container Structure And Fit-Out	Verify Container Structure And Fit-Out requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Sensing And Instrumentation	Review Sensing And Instrumentation architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Control Compute And Communication	Review Control Compute And Communication architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Control Compute And Communication	Verify Control Compute And Communication requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Safety Protection And Interlocks	Review Safety Protection And Interlocks	verification_intake	publishable	Collect reviewer evidence through the

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
				architecture allocation			verification intake template.
verification_evidence	major	ready for intake	Environmental Interface	Review Environmental Interface architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	DC And AC Power Distribution	Review DC And AC Power Distribution architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	DC And AC Power Distribution	Verify DC And AC Power Distribution requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Maintenance And Serviceability	Review Maintenance And Serviceability architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Maintenance And Serviceability	Verify Maintenance And Serviceability requirement calculations	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Coolant Circulation Process	Review Coolant Circulation Process architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Operator HMI And Labelling	Review Operator HMI And Labelling architecture allocation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Energy Storage Source -> Power	Verify dc_bus interface contract	verification_intake	publishable	Collect reviewer evidence through the

Kind	Priority	Status	Scope	Gap	Closure Path	Blocks	Required Evidence
			Conversion System				verification intake template.
verification_evidence	major	ready for intake	Energy Storage Source -> Environmental Interface	Verify thermal_loop interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	Energy Storage Source -> Control Compute And Communication	Verify bms_network interface contract	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for IEC-62619	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for UL-9540A	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Compliance	Plan evidence for G99	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for Thermal runaway propagation	verification_intake	publishable	Collect reviewer evidence through the verification intake template.
verification_evidence	major	ready for intake	System Risk Register	Review risk control for DC arc fault	verification_intake	publishable	Collect reviewer evidence through the verification intake template.

Closure Plan

4 active phases. Next phase: sourcing_intake. Rows by phase: architecture 0, sourcing 9, engineering 8, verification 27, score repair 1.

#	Phase	Rows	B/M/m	Blocks	Rationale	Exit Criteria	Top Gap IDs
2	Admit Source-Backed BoM Evidence	9	9/0/0	bom; publishable	Critical BoM costs, suppliers, manufacturers and MPNs must be source-backed before cost claims are credible.	Every critical BoM line has admitted supplier, manufacturer, MPN, unit cost and source reference. BoM provenance manifest has zero critical missing-source claims.	assumption:metric:capex_gbp; assumption:metric:opex_annual_gbp; readiness:2; sourcing:bms_master_controller; sourcing:dc_contactor_or_breaker; sourcing:fire_detection_suppression; sourcing:lfp_prismatic_cells; sourcing:pcs_inverter; plus 1 more
3	Complete Engineering Review	8	0/8/0	publishable	Deterministic arithmetic can expose claims, but engineering judgment must review envelopes, assumptions and operating cases.	All calculation and assumption review rows are accepted, revised or explicitly deferred with reviewer rationale.	assumption:compliance:G99; assumption:compliance:IEC-62619; assumption:compliance:UL-9540A; assumption:metric:headline_output; assumption:sanity:bess_c_rate; assumption:sanity:bess_container_energy_density; assumption:sanity:bess_duration; calculation:bess_nameplate_annual_throughput_mwh
4	Accept Reviewer Evidence	27	2/25/0	publishable	Reviewer acceptance is tracked separately from deterministic scoring and must close before publication.	Every non-source verification activity is accepted, or explicitly deferred and removed from publication criteria.	readiness:3; readiness:4; verification:calculation:control_communication; verification:calculation:energy_conversion_transduction; verification:calculation:energy_storage_source;

#	Phase	Rows	B/M/m	Blocks	Rationale	Exit Criteria	Top Gap IDs
							verification:calculation:maintenance_serviceability; verification:calculation:power_distribution; verification:calculation:structure_containment; plus 12 more
5	Repair Remaining Section Scores	1	1/0/0	publishable	After evidence intake, rerun scoring to remove residual deterministic section blockers.	Every scored report section is at or above target and readiness verdict is publishable.	readiness:1

Depth Benchmark

chain-v2-adapted numeric analysis only. Benchmark uses aggregate counts only. Chain-v2 component names, prose, design content, prices and part numbers are not imported into the scratch generator.

2 meeting benchmark, 5 below benchmark, average comparable ratio 0.83.

Dimension	Scratch	Benchmark	Ratio	Status	Notes	Action
Functional modules	11 modules	11 modules	1	meets	Core architecture breadth.	Add missing functional modules only when the product class genuinely requires them.
Submodules	46 submodules	48 submodules	0.96	below	Submodule count is a rough proxy for engineering decomposition depth.	Expand thin modules with concrete sub-functions and carrier interfaces.
Component candidates	204 components	241 components	0.85	below	Component-word count is a rough proxy for BoM review surface, not sourcing quality.	Add justified component candidates where submodules still hide assemblies.
Interface link volume	50 links	54 links	0.93	below	Compares link volume only; chain-v2 grammar links and scratch interface edges are not semantically identical.	Add interface contracts only where they clarify a real energy, material, signal, load or service path.
Admitted priced-line ratio	0 ratio	0.86 ratio	0	below	Chain-v2 priced words are treated as benchmark coverage only; scratch admits prices only with explicit sourcing evidence.	Use sourcing intake to admit source-backed price/manufacturer/MPN evidence; do not copy benchmark estimates.
Sections scoring >=8	5 sections	6 sections	0.83	below	Readiness gate requires every scored section to reach the target.	Clear section-specific readiness actions, especially BoM evidence.
Verification activities	30 activities	24 activities	1.25	meets	Scratch-only evidence workflow depth; not compared to chain-v2 content.	Keep adding verification activities when new requirements, interfaces or standards are introduced.

Brief Requirements

Design a containerised 3.5 MWh battery energy storage system with 1 MW PCS, 28 tonne gross mass limit, and LFP prismatic cells.

Requirement	Value	Unit
Energy capacity	3.5	MWh
Power rating	1	MW
Mass constraint	28000	kg

Requirement Traceability

Requirement	Status	Value	Architecture Coverage	Evaluators	Notes
Energy capacity	covered	3.5 MWh	Energy Storage Source / Cell string; Power Conversion System / PCS inverter bridge; DC And AC Power Distribution / Main DC busbars	metric:headline_output; sanity:bess_c_rate; sanity:bess_duration; sanity:bess_container_energy_density	Energy capacity is traced through storage, conversion, distribution and sanity checks. Requirement has architecture coverage plus metric or engineering sanity coverage.
Power rating	covered	1 MW	Power Conversion System / PCS inverter bridge; DC And AC Power Distribution / Main DC busbars; Control Compute And Communication / BMS master controller	sanity:bess_c_rate; sanity:bess_duration	Power rating is traced through PCS, distribution and power-to-energy sanity checks. Requirement has architecture coverage plus metric or engineering sanity coverage.
Mass constraint	covered	28000 kg	Container Structure And Fit-Out / ISO container shell; Maintenance And Serviceability / Access doors and hardware	sanity:bess_container_energy_density	Mass is traced through structure and system-level energy-density sanity checks. Requirement has architecture coverage plus metric or engineering sanity coverage.

Requirement Coverage Gate

coverage_review_ready

3 requirement row(s): 0 accepted with evidence, 3 coverage-ready, 0 needing review, 0 blocked, 0 unlinked. Structural coverage ratio 1; accepted evidence ratio 0.

Requirement	Status	Present Signals	Missing Signals	Modules	Verification Activities	Required Action
Energy capacity	review ready	architecture_link; submodule_link; component_link; calculation_link; review_question; verification_activity	accepted_verification	energy_storage_source; energy_conversion_transduction; power_distribution	design_review:energy_storage_source; calculation:energy_storage_source; design_review:energy_conversion_transduction; calculation:energy_conversion_transduction; design_review:power_distribution; calculation:power_distribution	Collect accepted reviewer evidence; structural requirement coverage is ready for review.
Mass constraint	review ready	architecture_link; submodule_link; component_link; calculation_link; review_question; verification_activity	accepted_verification	structure_containment; maintenance_serviceability	design_review:structure_containment; calculation:structure_containment; design_review:maintenance_serviceability; calculation:maintenance_serviceability	Collect accepted reviewer evidence; structural requirement coverage is ready for review.
Power rating	review ready	architecture_link; submodule_link; component_link; calculation_link; review_question; verification_activity	accepted_verification	energy_conversion_transduction; power_distribution; control_computer_communication	design_review:energy_conversion_transduction; calculation:energy_conversion_transduction; design_review:control_computer_communication; calculation:control_computer_communication; design_review:power_distribution; calculation:power_distribution	Collect accepted reviewer evidence; structural requirement coverage is ready for review.

Engineering Assurance Matrix

3 requirements: 0 accepted, 0 ready, 3 needing review, 0 blocked, 0 unlinked. 3 have calculation links, 3 have review questions, 0 have accepted verification evidence.

Requirement	Overall	Architecture	Modules	Calculations	Review Questions	Verification	Blockers	Next Action
Energy capacity	needs review	covered	Energy Storage Source; Power Conversion System; DC And AC Power Distribution	bess_discharge_duration_h; bess_system_energy_density_wh_per_kg; bess_nameplate_annual_throughput_mwh	assumption_resolution: component: energy_conversion_version_traction_plus_inverter_plus_cs_inverter; assumption_resolution: component: energy_storage_source_cell_string_lfp_prismatic_cells; assumption_resolution: component: energy_storage_source_dc_string_output_dc_contactor_or_breaker; assumption_resolution: sanity:bess_charge_rate; assumption_resolution: sanity:bess_container_energy_density; plus 24 more	pending; pending; pending; pending	none	Complete engineering review questions, calculations or deferred verification evidence.
Mass constraint	needs review	covered	Container Structure And Fit-Out; Maintenance And Serviceability	bess_system_energy_density_wh_per_kg	assumption_resolution: sanity:bess_container_energy_density; calculation_envelope:bess_system_energy_density_wh_per_kg; module_allocation:maintenance_serviceability; module_allocation:structure_containment; submodule_allocation:maintenance_serviceability;access_dors; plus 7 more	pending; pending; pending	none	Complete engineering review questions, calculations or deferred verification evidence.

Requirement	Overall	Architecture	Modules	Calculations	Review Questions	Verification	Blockers	Next Action
Power rating	needs review	covered	Power Conversion System; DC And AC Power Distribution; Control Compute And Communication	bess_discharge_duration_h	assumption_resolution: component: control_compute_communication_bms_master_controller; assumption_resolution: component: energy_conversion_transduction_plus_inverter_plus_cs_inverter; assumption_resolution: sanity:bess_c_rate; assumption_resolution: sanity:bess_container_energy_density; assumption_resolution: sanity:bess_duration; plus 20 more	pending; pending; pending; pending; pending	none	Complete engineering review questions, calculations or deferred verification evidence.

Compiler Stage Trace

Stage	Status	Summary	Metrics	Evidence	Limitations
Brief Parsing	passed	Extracted quantified requirements from the user brief before design generation.	requirement_count: 3; numeric_fact_count : 3; requirement_trace_covered: 3; requirement_trace_partial: 0; requirement_trace_uncovered: 0	capacity_mwh: 3.5 MWh; power_mw: 1 MW; mass_kg: 28000 kg; capacity_mwh: covered; power_mw: covered; mass_kg: covered	Parsing is deterministic and currently recognises only a small set of hardware requirement patterns.
Product-Class Selection	passed	Selected energy_storage with high confidence from brief keywords.	selected_class: energy_storage; confidence: high; keyword_scores: energy_storage:6	energy_storage: 6	Classifier is keyword based; ambiguous multi-domain briefs need an LLM or richer taxonomy pass.
Universal Module Architecture	passed	Built a 11-module energy_storage architecture from the scratch universal architecture grammar.	module_count: 11; architecture_source : scratch_universal_architecture	Energy Storage Source; Power Conversion System; Container Structure And Fit-Out; Sensing And Instrumentation; Control Compute And Communication; Safety Protection And Interlocks; plus 5 more	The scratch grammar is hand-authored and deterministic; it still needs independent engineering review.
Submodule Expansion	passed	Expanded each module into engineering submodules with local purpose, interfaces and component candidates.	submodule_count: 46; average_submodules_per_module: 4.2	energy_storage_source.cell_string; energy_storage_source.rack_structure; energy_storage_source.bms_slave; energy_storage_source.dc_string_output; energy_storage_source.module_interconnect_harness; energy_storage_source.rack_thermal_interface; plus 40 more	Submodule completeness is judged by deterministic coverage checks, not by a physics simulator.
Interface Graph	passed	Checked class-required module-to-module interfaces before allowing BoM review.	required_links: 3; present_links: 3; missing_links: 0; graph_nodes: 57; graph_edges: 96; shared_interface_edges: 47; missing_required_graph_edges: 0	energy_storage_source -> energy_conversion_transduction via dc_bus: present; energy_storage_source -> environmental_interface via thermal_loop: present; energy_storage_source ->	Interfaces prove allocation coverage only; they do not yet validate ratings, thermal margins or dynamic behaviour.

Stage	Status	Summary	Metrics	Evidence	Limitations
Component Candidates	passed	Converted allocated component words into a candidate BoM without supplier prices.	candidate_lines: 204; critical_candidate_lines: 6; admitted_price_lines: 0	control_compute_communication via bms_network: present lfp_prismatic_cells: LFP prismatic cells; cell_to_cell_busbar: cell-to-cell busbar; cell_terminal_hardware_set: cell terminal hardware set; cell_insulation_pad: cell insulation pad; module_fuse: module fuse; module_steel_frame: module steel frame; plus 24 more	Showing first 30 of 204 component candidates in the stage trace.
Architecture Readiness Gate	passed	Architecture validators found no blocker or major issue, so candidate BoM review can start.	ready_for_bom: true; blocking_issue_count: 0; sanity_pass: 3; sanity_warn: 0; sanity_fail: 0	pass BESS C-Rate: 0.29 C; pass Rated Duration: 3.5 hours; pass Container-Level Energy Density: 125 Wh/kg	Passing this gate means deterministic architecture checks passed; it is not a certified engineering sign-off.
Sourcing And BoM Admission	warning	No supplier, manufacturer, part number or cost claims have been admitted because no source-backed evidence was provided.	admission_status: not_started; candidate_lines: 204; admitted_lines: 0; unpriced_lines: 204; unpriced_critical_lines: 6; rejected_source_records: 0	none	BoM cost remains intentionally blocked until 6 critical candidate line(s) have explicit source evidence.

Engineering Architecture Review

Ready for BoM review

11 modules, 46 sub-modules, 204 allocated component words.

From	To	Interface	Status	Engineering Reason
energy_storage_source	energy_conversion_transduction	dc_bus	OK	Battery racks must feed the PCS over a declared DC bus.
energy_storage_source	environmental_interface	thermal_loop	OK	Cell heat must be coupled to the thermal management loop.
energy_storage_source	control_compute_communication	bms_network	OK	Cell/rack state must be visible to BMS and EMS supervision.

Interface Graph Summary

Module Nodes	Submodule Nodes	Shared Interface Edges	Required Edges	Missing Required Edges
11	46	47	3	0

Interface Contract Matrix

3/3 required contracts present. 16 shared interface families, 0 local-only declarations.

Interface	From Module	From Carrier	To Module	To Carrier	Status	Reason	Notes
dc_bus	Energy Storage Source	Cell string; DC string output	Power Conversion System	PCS inverter bridge	present	Battery racks must feed the PCS over a declared DC bus.	Both endpoints expose the required interface through at least one submodule carrier.
thermal_loop	Energy Storage Source	Cell string; Rack thermal interface	Environmental Interface	HVAC unit; Ventilation path; Heat rejection coil	present	Cell heat must be coupled to the thermal management loop.	Both endpoints expose the required interface through at least one submodule carrier.
bms_network	Energy Storage Source	BMS slave board	Control Compute And Communication	BMS master controller	present	Cell/rack state must be visible to BMS and EMS supervision.	Both endpoints expose the required interface through at least one submodule carrier.

No local-only interface declarations in the current cross-module index.

Interface Verification Gate

interface_review_ready

3 required interface row(s): 0 accepted, 3 review-ready, 0 pending evidence, 0 deferred, 0 blocked. Structural pass ratio 1; accepted evidence ratio 0.

Interface	From	To	Carrier Status	Verification Activity	Ledger	Status	Required Action
dc_bus	Energy Storage Source	Power Conversion System	both_endpoints_carried	interface_review:energy_storage_source:energy_conversion_transduction:dc_bus	pending	review_ready	Collect named interface-review evidence through verification intake.
thermal_loop	Energy Storage Source	Environmental Interface	both_endpoints_carried	interface_review:energy_storage_source:environmental_interface:thermal_loop	pending	review_ready	Collect named interface-review evidence through verification intake.
bms_network	Energy Storage Source	Control Compute And Communication	both_endpoints_carried	interface_review:energy_storage_source:control_compute_communication:bms_network	pending	review_ready	Collect named interface-review evidence through verification intake.

Component Allocation Gate

allocation_ready

46/46 submodules ready; 0 review, 0 blocked. Component words 204; critical parts 6/6; duplicate groups 0. Allocation ratio 1; critical allocation ratio 1.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Energy Storage Source	Cell string	ready	5	dc_bus; thermal_loop; bms_tap_harness	lfp_prismatic_cells	none	Submodule has component candidates and no allocation blocker.
Energy Storage Source	Rack structure	ready	5	mechanical_mounts; service_accesses	none	none	Submodule has component candidates and no allocation blocker.
Energy Storage Source	BMS slave board	ready	5	bms_network; sensor_harnesses	none	none	Submodule has component candidates and no allocation blocker.
Energy Storage Source	DC string output	ready	5	dc_bus; hardwired_trip	dc_contactor_or_breaker	none	Submodule has component candidates and no allocation blocker.
Energy Storage Source	Module interconnect harness	ready	4	bms_tap_harness; sensor_harnesses; service_accesses	none	none	Submodule has component candidates and no allocation blocker.
Energy Storage Source	Rack thermal interface	ready	5	thermal_loop; mechanical_mounts; service_accesses	none	none	Submodule has component candidates and no allocation blocker.
Power Conversion System	PCS inverter bridge	ready	6	dc_bus; ac_bus; modbus_tcp	pcs_inverter	none	Submodule has component candidates and no allocation blocker.
Power Conversion System	Transformer stage	ready	5	ac_bus; thermal_path	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Power Conversion System	Grid filter	ready	4	ac_bus; protective_earth	none	none	Submodule has component candidates and no allocation blocker.
Power Conversion System	Grid protection relay	ready	5	ac_bus; modbus_tcp; hardwired_trip; protective_earth	none	none	Submodule has component candidates and no allocation blocker.
Container Structure And Fit-Out	ISO container shell	ready	4	lifting_points; service_accesses	none	none	Submodule has component candidates and no allocation blocker.
Container Structure And Fit-Out	Internal partitions	ready	4	fire_boundary ; service_accesses	none	none	Submodule has component candidates and no allocation blocker.
Container Structure And Fit-Out	Rack mounting rails	ready	4	mechanical_mounts	none	none	Submodule has component candidates and no allocation blocker.
Container Structure And Fit-Out	Cable penetration sealing	ready	5	fire_boundary ; service_accesses; protective_earth	none	none	Submodule has component candidates and no allocation blocker.
Sensing And Instrumentation	Electrical measurement	ready	4	sensor_bus; dc_bus	none	none	Submodule has component candidates and no allocation blocker.
Sensing And Instrumentation	Environmental sensors	ready	4	sensor_bus; alarm_bus	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Sensing And Instrumentation	Thermal sensors	ready	4	sensor_bus; thermal_loop	none	none	Submodule has component candidates and no allocation blocker.
Sensing And Instrumentation	Event recording	ready	5	sensor_bus; alarm_bus; site_scada	none	none	Submodule has component candidates and no allocation blocker.
Control Compute And Communication	BMS master controller	ready	4	bms_network; hardwired_trip	bms_master_controller	none	Submodule has component candidates and no allocation blocker.
Control Compute And Communication	EMS controller	ready	4	modbus_tcp; site_scada	none	none	Submodule has component candidates and no allocation blocker.
Control Compute And Communication	Communications gateway	ready	4	site_scada; modbus_tcp	none	none	Submodule has component candidates and no allocation blocker.
Control Compute And Communication	Cybersecurity monitoring	ready	5	site_scada; modbus_tcp; service_access	none	none	Submodule has component candidates and no allocation blocker.
Safety Protection And Interlocks	Fire suppression system	ready	5	alarm_bus; fire_boundary	fire_detection_suppression	none	Submodule has component candidates and no allocation blocker.
Safety Protection And Interlocks	Emergency stop chain	ready	4	hardwired_trip; dc_bus; ac_bus	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Safety Protection And Interlocks	Surge and fault protection	ready	4	dc_bus; ac_bus; protective_earth	none	none	Submodule has component candidates and no allocation blocker.
Safety Protection And Interlocks	Deflagration venting	ready	5	fire_boundary ; alarm_bus; service_access	none	none	Submodule has component candidates and no allocation blocker.
Environmental Interface	HVAC unit	ready	4	electrical_power; thermal_loop; thermal_path; sensor_bus	none	none	Submodule has component candidates and no allocation blocker.
Environmental Interface	Ventilation path	ready	4	thermal_loop; thermal_path; service_access	none	none	Submodule has component candidates and no allocation blocker.
Environmental Interface	Thermal insulation	ready	4	fire_boundary ; thermal_path	none	none	Submodule has component candidates and no allocation blocker.
Environmental Interface	Heat rejection coil	ready	5	thermal_loop; thermal_path; electrical_power; service_access	none	none	Submodule has component candidates and no allocation blocker.
DC And AC Power Distribution	Main DC busbars	ready	4	dc_bus; service_access	none	none	Submodule has component candidates and no allocation blocker.
DC And AC Power Distribution	AC switchgear	ready	4	ac_bus; protective_earth; hardwired_trip	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
DC And AC Power Distribution	Auxiliary power distribution	ready	4	electrical_power; protective_earth	none	none	Submodule has component candidates and no allocation blocker.
DC And AC Power Distribution	Earthing and lightning protection	ready	5	protective_earth; service_accesses	none	none	Submodule has component candidates and no allocation blocker.
Maintenance And Serviceability	Access doors and hardware	ready	4	service_accesses	none	none	Submodule has component candidates and no allocation blocker.
Maintenance And Serviceability	Service lighting	ready	4	electrical_power; service_accesses	none	none	Submodule has component candidates and no allocation blocker.
Maintenance And Serviceability	Lifting and handling points	ready	4	lifting_points; mechanical_mounts	none	none	Submodule has component candidates and no allocation blocker.
Maintenance And Serviceability	Commissioning ports	ready	5	service_accesses; sensor_bus; dc_bus	none	none	Submodule has component candidates and no allocation blocker.
Coolant Circulation Process	Coolant pump station	ready	5	thermal_loop; electrical_power; sensor_bus	thermal_management_loop	none	Submodule has component candidates and no allocation blocker.
Coolant Circulation Process	Filtration and treatment	ready	4	thermal_loop; service_accesses	none	none	Submodule has component candidates and no allocation blocker.

Module	Submodule	Status	Components	Interfaces	Critical Components	Duplicate IDs	Required Action
Coolant Circulation Process	Expansion and air removal	ready	4	thermal_loop; sensor_bus	none	none	Submodule has component candidates and no allocation blocker.
Coolant Circulation Process	Heat exchanger manifold	ready	5	thermal_loop; service_access; sensor_bus	none	none	Submodule has component candidates and no allocation blocker.
Operator HMI And Labelling	Local HMI panel	ready	4	site_scada; service_access	none	none	Submodule has component candidates and no allocation blocker.
Operator HMI And Labelling	External labelling	ready	4	service_access	none	none	Submodule has component candidates and no allocation blocker.
Operator HMI And Labelling	Operator workflow aids	ready	4	service_access; alarm_bus	none	none	Submodule has component candidates and no allocation blocker.
Operator HMI And Labelling	Remote notification interface	ready	5	site_scada; alarm_bus; service_access	none	none	Submodule has component candidates and no allocation blocker.

Submodule Engineering Gate

submodule_engineering_review_required

0/46 submodules pass; 46 review, 0 blocked. Component words 204; linked requirements 8; carrier contracts 10/10; local-only interfaces 101. Review questions accepted 0/102; verification accepted 0/68; critical unpriced lines 6.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Energy Storage Source	Cell string	review	5	3	1	2/2	0/4	0/5	1	0.6667	Review 1 local-only interface declaration(s): bms_tapp_harness. 4 engineering review question(s) await acceptance. 4 verification activity row(s) await acceptance. 1 evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Energy Storage Source	Rack structure	review	5	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): mechanical_mounts, service_access. 2 engineering review question(s) await acceptance. 1 verification

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											ion activity row(s) await accepted evidence.
Energy Storage Source	BMS slave board	review	5	2	0	1/1	0/3	0/2	0	0.6667	Review 1 local-only interface declaration(s): sensor_harness. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Energy Storage Source	DC string output	review	5	2	0	1/1	0/3	0/3	1	0.6667	Review 1 local-only interface declaration(s): hardwired_trip. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence. 1

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Energy Storage Source	Module interconnect harness	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): bms_tap_harness, sensor_harness, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Energy Storage Source	Rack thermal interface	review	5	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): mechanical_mounts, service_access. 3 engineering review

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Power Conversion System	PCS inverter bridge	review	6	3	2	1/1	0/3	0/4	1	0.6667	Review 2 local-only interface declaration(s): ac_bus, modbus_tcp. 3 engineering review question(s) await acceptance. 3 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufactur er/MPN evidence.
Power Conversion System	Transformer stage	review	5	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): ac_bus,

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											thermal_path. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Power Conversion System	Grid filter	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): ac_bus, protective_earth. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Power Conversion System	Grid protection relay	review	5	4	0	0/0	0/2	0/1	0	0.6667	Review 4 local-only interface declaration(s): ac_bus, modbus_tcp, hardwired_trip, protective_earth. 2

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Container Structure And Fit-Out	ISO container shell	review	4	2	1	0/0	0/2	0/2	0	0.6667	Review 2 local-only interface declaration(s): lifting_points, service_access. 2 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Container Structure And Fit-Out	Internal partitions	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): fire_boundary, service_access. 2 engineering review question(s)

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											n(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Container Structure And Fit-Out	Rack mounting rails	review	4	1	0	0/0	0/2	0/1	0	0.6667	Review 1 local-only interface declaration(s): mechanical_mounts. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Container Structure And Fit-Out	Cable penetration sealing	review	5	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): fire_boundary, service_access, protective_earth. 2 engineering review question(s) await acceptance. 1

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											verification activity row(s) await acceptance evidence.
Sensing And Instrumentation	Electrical measurement	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): sensor_bus, dc_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await acceptance evidence.
Sensing And Instrumentation	Environmental sensors	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): sensor_bus, alarm_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await acceptance

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											d evidence.
Sensing And Instrumentation	Thermal sensors	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): sensor_bus, thermal_loop. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Sensing And Instrumentation	Event recording	review	5	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): sensor_bus, alarm_bus, site_scada. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Control Computer And Communication	BMS master controller	review	4	2	1	1/1	0/3	0/4	1	0.6667	Review 1 local-only interface declaration(s): hardwired_trip. 3 engineering review question(s) await acceptance. 3 verification activity row(s) await acceptance. 1 evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Control Computer And Communication	EMS controller	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): modbus_tcp, site_scada. 2 engineering review question(s) await acceptance. 1 verification activity

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											row(s) await acceptance. evidence.
Control Computer And Communication	Communications gateway	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): site_scada, modbus_tcp. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await acceptance. evidence.
Control Computer And Communication	Cybersecurity monitoring	review	5	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): site_scada, modbus_tcp, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await acceptance.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											d evidence.
Safety Protection And Interlocks	Fire suppression system	review	5	2	0	0/0	0/2	0/2	1	0.6667	Review 2 local-only interface declaration(s): alarm_bus, fire_boundary. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/manufacturer/MPN evidence.
Safety Protection And Interlocks	Emergency stop chain	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): hardwired_trip, dc_bus, ac_bus. 2 engineering review question(s) await

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											acceptance. 1 verification activity row(s) await accepted evidence.
Safety Protection And Interlocks	Surge and fault protection	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): dc_bus, ac_bus, protective_earth. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Safety Protection And Interlocks	Deflagration venting	review	5	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): fire_boundary, alarm_bus, service_access. 2 engineering review question(s) await acceptance. 1

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											verification activity row(s) await acceptance evidence.
Environmental Interface	HVAC unit	review	4	4	0	1/1	0/3	0/2	0	0.6667	Review 3 local-only interface declaration(s): electrical_power, thermal_path, sensor_bus. 3 engineering review question(s) await acceptance. 2 verification activity row(s) await acceptance evidence.
Environmental Interface	Ventilation path	review	4	3	0	1/1	0/3	0/2	0	0.6667	Review 2 local-only interface declaration(s): thermal_path, service_access. 3 engineering review question(s) await acceptance. 2 verification

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											activity row(s) await accepted evidence.
Environmental Interface	Thermal insulation	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): fire_boundary, thermal_path. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Environmental Interface	Heat rejection coil	review	5	4	0	1/1	0/3	0/2	0	0.6667	Review 3 local-only interface declaration(s): thermal_path, electrical_power, service_access. 3 engineering review question(s) await acceptance. 2 verification activity row(s)

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											await accepted evidence.
DC And AC Power Distribution	Main DC busbars	review	4	2	2	0/0	0/2	0/2	0	0.6667	Review 2 local-only interface declaration(s): dc_bus, service_access. 2 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
DC And AC Power Distribution	AC switchgear	review	4	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): ac_bus, protective_earth, hardwired_trip. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											evidence.
DC And AC Power Distribution	Auxiliary power distribution	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): electrical_power, protective_earth. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
DC And AC Power Distribution	Earthing and lightning protection	review	5	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): protective_earth, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Maintenance And Serviceability	Access doors and hardware	review	4	1	1	0/0	0/2	0/2	0	0.6667	Review 1 local-only interface declaration(s): service_access. 2 engineering review question(s) await acceptance. 2 verification activity row(s) await accepted evidence.
Maintenance And	Service lighting	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Serviceability											electrical_power, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Maintenance And Serviceability	Lifting and handling points	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): lifting_points, mechanical_mounts. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Maintenance And Serviceability	Commissioning ports	review	5	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s):

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											service_access, sensor_bus, dc_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Coolant Circulation Process	Coolant pump station	review	5	3	0	0/0	0/2	0/2	1	0.6667	Review 3 local-only interface declaration(s): thermal_loop, electrical_power, sensor_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence. 1 critical component line(s) still need source-backed cost/ma

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											manufacturer/MPN evidence.
Coolant Circulation Process	Filtration and treatment	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): thermal_loop, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Coolant Circulation Process	Expansion and air removal	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): thermal_loop, sensor_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Coolant Circulation Process	Heat exchanger manifold	review	5	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declaration(s): thermal_loop, service_access, sensor_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Operator HMI And Labelling	Local HMI panel	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): site_scada, service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
Operator HMI And Labelling	External labelling	review	4	1	0	0/0	0/2	0/1	0	0.6667	Review 1 local-only interface declaration(s): service_access. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Operator HMI And Labelling	Operator workflow aids	review	4	2	0	0/0	0/2	0/1	0	0.6667	Review 2 local-only interface declaration(s): service_access, alarm_bus. 2 engineering review question(s) await acceptance. 1 verification activity row(s) await accepted evidence.
Operator HMI And Labelling	Remote notification interface	review	5	3	0	0/0	0/2	0/1	0	0.6667	Review 3 local-only interface declarat

Module	Submodule	Verdict	Components	Interfaces	Req Links	Carrier Contracts	Review Qs	Verification	Critical Unpriced	Pass Ratio	Required Action
											ion(s): site_sca da, alarm_b us, service_ access. 2 engineer ing review questio n(s) await accepta nce. 1 verificat ion activity row(s) await accepte d evidenc e.

Module Engineering Gate

module_engineering_review_required

0/11 modules pass; 11 review, 0 blocked. Submodules/components: 46/204. Interface carriers 6/6; review questions accepted 0/94; critical unpriced lines 6 across 5 module(s).

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
Energy Storage Source	review	6	29	1	3/3	0/18	2	0.86	2 critical BoM line(s) need source-backed cost evidence. 18 engineering review question(s) await acceptance. 8 assumption row(s) need review or source evidence.
Power Conversion System	review	4	20	2	1/1	0/13	1	0.86	1 critical BoM line(s) need source-backed cost evidence. 13 engineering review question(s) await acceptance. 5 assumption row(s) need review or source evidence.
Container Structure And Fit-Out	review	4	17	1	0/0	0/7	0	0.86	7 engineering review question(s) await acceptance. 1 assumption row(s) need review or source evidence.
Sensing And Instrumentation	review	4	17	0	0/0	0/5	0	0.71	5 engineering review question(s) await acceptance.

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
									e. 1 assumptio n row(s) need review or source evidence.
Control Compute And Communic ation	review	4	17	1	1/1	0/10	1	0.86	1 critical BoM line(s) need source- backed cost evidence. 10 engineerin g review question(s) await acceptanc e. 4 assumptio n row(s) need review or source evidence.
Safety Protection And Interlocks	review	4	18	0	0/0	0/6	1	0.71	1 critical BoM line(s) need source- backed cost evidence. 6 engineerin g review question(s) await acceptanc e. 2 assumptio n row(s) need review or source evidence.

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
Environmental Interface	review	4	17	0	1/1	0/6	0	0.71	6 engineering review question(s) await acceptance. 1 assumption row(s) need review or source evidence.
DC And AC Power Distribution	review	4	17	2	0/0	0/11	0	0.86	11 engineering review question(s) await acceptance. 4 assumption row(s) need review or source evidence.
Maintenance And Serviceability	review	4	17	1	0/0	0/7	0	0.86	7 engineering review question(s) await acceptance. 2 assumption row(s) need review or source evidence.
Coolant Circulation Process	review	4	18	0	0/0	0/6	1	0.71	1 critical BoM line(s) need source-backed cost evidence. 6 engineering review question(s) await acceptance. 2 assumption row(s) need review or source evidence.

Module	Verdict	Submodul es	Compone nts	Req Links	Interface s	Review Qs	Critical Unpriced	Pass Ratio	Required Action
Operator HMI And Labelling	review	4	17	0	0/0	0/5	0	0.71	5 engineerin g review question(s) await acceptanc e.

Module Review Roll-Up

6 ready, 5 sourcing-blocked, 0 needing attention.

Module	Status	Submodules	Components	Required Edges	Requirements	Critical Sourcing Blockers	Notes
Energy Storage Source	sourcing	6	29	3/3	capacity_mwh	lfp_prismatic_cells; dc_contactor_or_breaker	2 critical line(s) need source-backed evidence before BoM pass.
Power Conversion System	sourcing	4	20	1/1	capacity_mwh ; power_mw	pcs_inverter	1 critical line(s) need source-backed evidence before BoM pass.
Container Structure And Fit-Out	ready	4	17	0/0	mass_kg	none	No critical blockers; 17 candidate line(s) remain for later sourcing.
Sensing And Instrumentation	ready	4	17	0/0	none	none	No critical blockers; 17 candidate line(s) remain for later sourcing.
Control Compute And Communication	sourcing	4	17	1/1	power_mw	bms_master_controller	1 critical line(s) need source-backed evidence before BoM pass.
Safety Protection And Interlocks	sourcing	4	18	0/0	none	fire_detection_suppression	1 critical line(s) need source-backed evidence before BoM pass.
Environmental Interface	ready	4	17	1/1	none	none	No critical blockers; 17 candidate line(s) remain for later sourcing.
DC And AC Power Distribution	ready	4	17	0/0	capacity_mwh ; power_mw	none	No critical blockers; 17 candidate line(s) remain for later sourcing.

Module	Status	Submodules	Components	Required Edges	Requirements	Critical Sourcing Blockers	Notes
Maintenance And Serviceability	ready	4	17	0/0	mass_kg	none	No critical blockers; 17 candidate line(s) remain for later sourcing.
Coolant Circulation Process	sourcing	4	18	0/0	none	thermal_management_loop	1 critical line(s) need source-backed evidence before BoM pass.
Operator HMI And Labelling	ready	4	17	0/0	none	none	No critical blockers; 17 candidate line(s) remain for later sourcing.

Engineering Review Pack

81 reviewer questions: 0 accepted, 62 ready, 19 needing review, 0 blocked. Coverage: 11 modules, 46 submodules, 3 interfaces, 3 calculations, 18 assumptions.

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
assumption_resolution	major	needs review	G99	Can the G99 assumption be accepted for architecture review, or must it be replaced with evidence?	Protection settings, relay test certificates and commissioning records.	none	none	none	none
assumption_resolution	major	needs review	IEC-62619	Can the IEC-62619 assumption be accepted for architecture review, or must it be replaced with evidence?	Cell certificates and pack-level safety validation.	none	none	none	none
assumption_resolution	major	needs review	UL-9540A	Can the UL-9540A assumption be accepted for architecture review, or must it be replaced with evidence?	Cell, module and unit-level propagation evidence.	none	none	none	none
assumption_resolution	major	needs review	BMS master controller	Can the BMS master controller assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	control_compute_communication	none	none	none
assumption_resolution	major	needs review	PCS inverter	Can the PCS inverter assumption be accepted	Admit supplier, manufacturer, MPN where applicable,	energy_conversion_transduction	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				for architecture review, or must it be replaced with evidence?	unit cost, lead time and evidence reference through sourcing intake.				
assumption_resolution	major	needs review	LFP prismatic cells	Can the LFP prismatic cells assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	energy_storage_source	none	none	none
assumption_resolution	major	needs review	DC contactor or breaker	Can the DC contactor or breaker assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	energy_storage_source	none	none	none
assumption_resolution	major	needs review	Thermal management loop	Can the Thermal management loop assumption be accepted for architecture review, or must it be replaced with evidence?	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.	mass_fluid_transport_process	none	none	none
assumption_resolution	major	needs review	Fire detection/suppression	Can the Fire detection/suppression assumption be accepted	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time	safety_protection	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				for architecture review, or must it be replaced with evidence?	and evidence reference through sourcing intake.				
assumption_resolution	major	needs review	energy_storage_source -> control_computing_communication	Can the energy_storage_source -> control_computing_communication assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	bms_network	none	none
assumption_resolution	major	needs review	energy_storage_source -> energy_conversion_transduction	Can the energy_storage_source -> energy_conversion_transduction assumption be accepted for architecture review, or must it be replaced with evidence?	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	dc_bus	none	none
assumption_resolution	major	needs review	energy_storage_source -> environmental_interface	Can the energy_storage_source -> environmental_interface assumption be accepted for architecture review, or must it be replaced	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.	none	thermal_load	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				with evidence?					
assumption_resolution	major	needs review	Estimated CAPEX	Can the Estimated CAPEX assumption be accepted for architecture review, or must it be replaced with evidence?	Admit source-backed BoM pricing before treating this cost metric as a claim.	none	none	none	none
assumption_resolution	major	needs review	Annual MWh delivered	Can the Annual MWh delivered assumption be accepted for architecture review, or must it be replaced with evidence?	Review the formula, input values and operating envelope before treating this metric as a performance claim.	none	none	none	none
assumption_resolution	major	needs review	Estimated annual OPEX	Can the Estimated annual OPEX assumption be accepted for architecture review, or must it be replaced with evidence?	Admit source-backed BoM pricing before treating this cost metric as a claim.	none	none	none	none
assumption_resolution	major	needs review	BESS C-Rate	Can the BESS C-Rate assumption be accepted for architecture review, or must it be replaced with evidence?	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.	energy_storage_source; energy_conversion_transduction; power_distribution; control_compute_communication	none	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
assumption_resolution	major	needs review	Container-Level Energy Density	Can the Container-Level Energy Density assumption be accepted for architecture review, or must it be replaced with evidence?	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.	energy_storage_source; energy_conversion_transduction; power_distribution; structure_containment; maintenance_serviceability	none	none	none
assumption_resolution	major	needs review	Rated Duration	Can the Rated Duration assumption be accepted for architecture review, or must it be replaced with evidence?	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.	energy_storage_source; energy_conversion_transduction; power_distribution; control_compute_communication	none	none	none
calculation_envelope	major	needs review	BESS annual nameplate throughput	Does the BESS annual nameplate throughput calculation support the architecture envelope, or does the concept need revision?	Confirm cycles per year, usable depth of discharge, round-trip efficiency and degradation assumptions.	energy_storage_source; energy_conversion_transduction; power_distribution	none	bess_nameplate_annual_throughput_mwh	none
calculation_envelope	major	ready	BESS discharge duration	Does the BESS discharge duration calculation support the architecture envelope, or does the concept	Confirm use case, PCS rating, usable energy window, C-rate limits and warranty throughput.	energy_storage_source; energy_conversion_transduction; power_distribution; control_compute_communication	none	bess_discharge_duration_h	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
				need revision?					
calculation_envelope	major	ready	BESS system energy density	Does the BESS system energy density calculation support the architecture envelope, or does the concept need revision?	Review cell mass, racks, enclosure, HVAC, fire protection, transformer/PCS boundary and shipping mass allowance.	energy_storage_source; energy_conversion_transduction; power_distribution; structure_containment; maintenance_serviceability	none	bess_system_energy_density_kwh_per_kg	none
interface_contract	major	ready	Energy Storage Source -> Control Compute And Communication via bms_network	Is the bms_network interface between Energy Storage Source and Control Compute And Communication correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	energy_storage_source; control_compute_communication	bms_network	none	Both endpoints expose the required interface through at least one submodule carrier.
interface_contract	major	ready	Energy Storage Source -> Power Conversion System via dc_bus	Is the dc_bus interface between Energy Storage Source and Power Conversion System correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	energy_storage_source; energy_conversion_transduction	dc_bus	none	Both endpoints expose the required interface through at least one submodule carrier.

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculations	Blockers
interface_contract	major	ready	Energy Storage Source -> Environmental Interface via thermal_loop	Is the thermal_loop interface between Energy Storage Source and Environmental Interface correctly owned and physically carried?	Interface-review evidence naming both endpoint owners, carrier submodules, capacity limits and any deferred checks.	energy_storage_source; environmental_interface	thermal_loop	none	Both endpoints expose the required interface through at least one submodule carrier.
module_allocation	major	ready	Control Compute And Communication	Does Control Compute And Communication have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	control_compute_communication	bms_network; modbus_tcp; sensor_buses; alarm_bus; site_scada; service_access; hardwired_trip	bess_container; bess_duration	none
module_allocation	major	ready	Power Conversion System	Does Power Conversion System have the right responsibility boundary, submodule split and interface ownership for this concept?	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	energy_conversion_transduction	dc_bus; ac_bus; modbus_tcp; thermal_path; hardwired_trip; protective_earth	bess_container; bess_duration; bess_container_energy_density	none
module_allocation	major	ready	Energy Storage Source	Does Energy Storage Source have the right responsibility boundary, submodule split and interface ownership	Named engineering reviewer sign-off, change request or deferral note against the module allocation.	energy_storage_source	dc_bus; thermal_loop; bms_network; mechanical_mounts; bms_tap_harness; service_access; sensor_harness;	bess_container; bess_duration; bess_container_energy_density	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				for this concept?			hardwired_trip		
module_all ocation	major	ready	Environme ntal Interface	Does Environme ntal Interface have the right responsibi lity boundary, submodul e split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	environme ntal_interf ace	thermal_lo op; thermal_p ath; electrical_ power; sensor_bu s; service_ac cess; fire_bound ary	none	none
module_all ocation	major	ready	Operator HMI And Labelling	Does Operator HMI And Labelling have the right responsibi lity boundary, submodul e split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	hmi_ergon omics	site_scada ; service_ac cess; alarm_bus	none	none
module_all ocation	major	ready	Maintenan ce And Serviceabi lity	Does Maintenan ce And Serviceabi lity have the right responsibi lity boundary, submodul e split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	maintenan ce_service ability	service_ac cess; lifting_poi nts; protective _earth; sensor_bu s; dc_bus; electrical_ power; mechanica l_mounts	bess_cont ainer_ener gy_density	none
module_all ocation	major	ready	Coolant Circulation Process	Does Coolant Circulation Process have the right responsibi lity boundary, submodul e split and interface ownership	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	mass_fluid _transport _process	thermal_lo op; sensor_bu s; service_ac cess; electrical_ power	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				for this concept?					
module_all ocation	major	ready	DC And AC Power Distributio n	Does DC And AC Power Distributio n have the right responsibi lity boundary, submodul e split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	power_dis tribution	dc_bus; ac_bus; electrical_ power; protective _earth; hardwired _trip; service_ac cess	bess_c_ra te; bess_dura tion; bess_cont ainer_ener gy_density	none
module_all ocation	major	ready	Safety Protection And Interlocks	Does Safety Protection And Interlocks have the right responsibi lity boundary, submodul e split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	safety_pro tection	alarm_bus ; hardwired _trip; fire_bound ary; dc_bus; ac_bus; service_ac cess; protective _earth	none	none
module_all ocation	major	ready	Sensing And Instrument ation	Does Sensing And Instrument ation have the right responsibi lity boundary, submodul e split and interface ownership for this concept?	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	sensing_in strumenta tion	sensor_bu s; alarm_bus ; bms_netw ork; site_scada ; dc_bus; thermal_lo op	none	none
module_all ocation	major	ready	Container Structure And Fit-Out	Does Container Structure And Fit-Out have the right responsibi lity boundary, submodul e split and interface ownership	Named engineerin g reviewer sign-off, change request or deferral note against the module allocation.	structure_ containme nt	mechanica l_mounts; service_ac cess; fire_bound ary; lifting_poi nts; protective _earth	bess_cont ainer_ener gy_density	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				for this concept?					
submodul e_allocation	major	ready	Control Compute And Communication / BMS master controller	Do the componen t candidate s and interfaces assigned to BMS master controller make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	control_co mpute_co mmunicati on	bms_netw ork; hardwired _trip	bess_c_ra te; bess_dura tion	none
submodul e_allocation	major	ready	Control Compute And Communication / Communications gateway	Do the componen t candidate s and interfaces assigned to Communications gateway make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	control_co mpute_co mmunicati on	site_scada ; modbus_t cp	bess_c_ra te; bess_dura tion	none
submodul e_allocation	major	ready	Control Compute And Communication / Cybersecu rity monitoring	Do the componen t candidate s and interfaces assigned to Cybersecu rity monitoring make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	control_co mpute_co mmunicati on	site_scada ; modbus_t cp; service_ac cess	bess_c_ra te; bess_dura tion	none
submodul e_allocation	major	ready	Control Compute And Communication / EMS controller	Do the componen t candidate s and interfaces assigned to EMS controller	Reviewer note confirming the submodul e role, componen t allocation	control_co mpute_co mmunicati on	modbus_t cp; site_scada	bess_c_ra te; bess_dura tion	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				make engineering sense for its stated purpose?	and any missing carriers or capacity checks.				
submodule_allocation	major	ready	Power Conversion System / Grid filter	Do the component candidates and interfaces assigned to Grid filter make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_conversion_transduction	ac_bus; protective_earth	bess_c_rate; bess_duration; bess_container_energy_density	none
submodule_allocation	major	ready	Power Conversion System / Grid protection relay	Do the component candidates and interfaces assigned to Grid protection relay make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_conversion_transduction	ac_bus; modbus_tcp; hardwired_trip; protective_earth	bess_c_rate; bess_duration; bess_container_energy_density	none
submodule_allocation	major	ready	Power Conversion System / PCS inverter bridge	Do the component candidates and interfaces assigned to PCS inverter bridge make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_conversion_transduction	dc_bus; ac_bus; modbus_tcp	bess_c_rate; bess_duration; bess_container_energy_density	none
submodule_allocation	major	ready	Power Conversion System / Transformer stage	Do the component candidates and interfaces assigned to Transformer stage	Reviewer note confirming the submodule role, component allocation and any	energy_conversion_transduction	ac_bus; thermal_path	bess_c_rate; bess_duration; bess_container_energy_density	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
				make engineerin g sense for its stated purpose?	missing carriers or capacity checks.				
submodul e_allocatio n	major	ready	Energy Storage Source / BMS slave board	Do the componen t candidate s and interfaces assigned to BMS slave board make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	energy_st orage_sou rce	bms_netw ork; sensor_ha rness	bess_c_ra te; bess_dura tion; bess_cont ainer_ener gy_density	none
submodul e_allocatio n	major	ready	Energy Storage Source / Cell string	Do the componen t candidate s and interfaces assigned to Cell string make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	energy_st orage_sou rce	dc_bus; thermal_lo op; bms_tap_ harness	bess_c_ra te; bess_dura tion; bess_cont ainer_ener gy_density	none
submodul e_allocatio n	major	ready	Energy Storage Source / DC string output	Do the componen t candidate s and interfaces assigned to DC string output make engineerin g sense for its stated purpose?	Reviewer note confirming the submodul e role, componen t allocation and any missing carriers or capacity checks.	energy_st orage_sou rce	dc_bus; hardwired _trip	bess_c_ra te; bess_dura tion; bess_cont ainer_ener gy_density	none
submodul e_allocatio n	major	ready	Energy Storage Source / Module interconne ct harness	Do the componen t candidate s and interfaces assigned to Module	Reviewer note confirming the submodul e role, componen t	energy_st orage_sou rce	bms_tap_ harness; sensor_ha rness; service_ac cess	bess_c_ra te; bess_dura tion; bess_cont ainer_ener gy_density	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
submodule_allocation	major	ready	Energy Storage / Rack structure	interconnect harness make engineering sense for its stated purpose?	allocation and any missing carriers or capacity checks.				
				Do the component candidate s and interfaces assigned to Rack structure make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_storage_source	mechanical_mounts; service_access	bess_c_rate; bess_duration; bess_container_energy_density	none
submodule_allocation	major	ready	Energy Storage Source / Rack thermal interface	Do the component candidate s and interfaces assigned to Rack thermal interface make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	energy_storage_source	thermal_loop; mechanical_mounts; service_access	bess_c_rate; bess_duration; bess_container_energy_density	none
submodule_allocation	major	ready	Environmental Interface / Heat rejection coil	Do the component candidate s and interfaces assigned to Heat rejection coil make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	environmental_interface	thermal_loop; thermal_path; electrical_power; service_access	none	none
submodule_allocation	major	ready	Environmental Interface / HVAC unit	Do the component candidate s and interfaces assigned	Reviewer note confirming the submodule role, component	environmental_interface	electrical_power; thermal_loop; thermal_path;	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodule_allocation	major	ready	Environmental Interface / Thermal insulation	to HVAC unit make engineering sense for its stated purpose?	t allocation and any missing carriers or capacity checks.		sensor_bu s		
				Do the component candidate s and interfaces assigned to Thermal insulation make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	environme ntal_interf ace	fire_bound ary; thermal_p ath	none	none
submodule_allocation	major	ready	Environmental Interface / Ventilation path	Do the component candidate s and interfaces assigned to Ventilation path make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	environme ntal_interf ace	thermal_lo op; thermal_p ath; service_ac cess	none	none
submodule_allocation	major	ready	Operator HMI And Labelling / External labelling	Do the component candidate s and interfaces assigned to External labelling make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	hmi_ergon omics	service_ac cess	none	none
submodule_allocation	major	ready	Operator HMI And Labelling / Local HMI panel	Do the component candidate s and interfaces assigned to Local HMI panel	Reviewer note confirming the submodule role, component allocation	hmi_ergon omics	site_scada ; service_ac cess	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
submodule_allocation	major	ready	Operator HMI And Labelling / Operator workflow aids	make engineering sense for its stated purpose?	and any missing carriers or capacity checks.	hmi_ergonomics	service_access; alarm_bus	none	none
				Do the component candidate s and interfaces assigned to Operator workflow aids make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.				
				Do the component candidate s and interfaces assigned to Remote notification interface make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.				
submodule_allocation	major	ready	Operator HMI And Labelling / Remote notification interface	Do the component candidate s and interfaces assigned to Remote notification interface make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	hmi_ergonomics	site_scada ; alarm_bus ; service_access	none	none
submodule_allocation	major	ready	Maintenance And Serviceability / Access doors and hardware	Do the component candidate s and interfaces assigned to Access doors and hardware make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	maintenance_serviceability	service_access	bess_container_energy_density	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodul e_allocation	major	ready	Maintenan ce And Serviceabi lity / Commissi oning ports	Do the componen t candidate s and interfaces assigned to Commissi oning ports make engineerin g sense for its stated purpose?	Reviewer note confirming the submodule role, componen t allocation and any missing carriers or capacity checks.	maintenan ce_service ability	service_ac cess; sensor_bu s; dc_bus	bess_cont ainer_ener gy_density	none
submodul e_allocation	major	ready	Maintenan ce And Serviceabi lity / Lifting and handling points	Do the componen t candidate s and interfaces assigned to Lifting and handling points make engineerin g sense for its stated purpose?	Reviewer note confirming the submodule role, componen t allocation and any missing carriers or capacity checks.	maintenan ce_service ability	lifting_poi nts; mechanica l_mounts	bess_cont ainer_ener gy_density	none
submodul e_allocation	major	ready	Maintenan ce And Serviceabi lity / Service lighting	Do the componen t candidate s and interfaces assigned to Service lighting make engineerin g sense for its stated purpose?	Reviewer note confirming the submodule role, componen t allocation and any missing carriers or capacity checks.	maintenan ce_service ability	electrical_ power; service_ac cess	bess_cont ainer_ener gy_density	none
submodul e_allocation	major	ready	Coolant Circulation Process / Expansion and air removal	Do the componen t candidate s and interfaces assigned to Expansion and air removal make	Reviewer note confirming the submodule role, componen t allocation and any missing carriers or	mass_fluid _transport _process	thermal_lo op; sensor_bu s	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
submodule_allocation	major	ready	Coolant Circulation Process / Filtration and treatment	engineering sense for its stated purpose?	capacity checks.				
				Do the component candidates and interfaces assigned to Filtration and treatment make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	mass_fluid_transport_process	thermal_loop; service_access	none	none
submodule_allocation	major	ready	Coolant Circulation Process / Heat exchanger manifold	Do the component candidates and interfaces assigned to Heat exchanger manifold make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	mass_fluid_transport_process	thermal_loop; service_access; sensor_buses	none	none
submodule_allocation	major	ready	Coolant Circulation Process / Coolant pump station	Do the component candidates and interfaces assigned to Coolant pump station make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	mass_fluid_transport_process	thermal_loop; electrical_power; sensor_buses	none	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interface s	Calculatio ns	Blockers
submodule_allocation	major	ready	DC And AC Power Distribution / AC switchgear	Do the component candidates and interfaces assigned to AC switchgear make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	power_distribution	ac_bus; protective_earth; hardwired_trip	bess_c_rate; bess_duration; bess_container_energy_density	none
submodule_allocation	major	ready	DC And AC Power Distribution / Auxiliary power distribution	Do the component candidates and interfaces assigned to Auxiliary power distribution make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	power_distribution	electrical_power; protective_earth	bess_c_rate; bess_duration; bess_container_energy_density	none
submodule_allocation	major	ready	DC And AC Power Distribution / Main DC busbars	Do the component candidates and interfaces assigned to Main DC busbars make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	power_distribution	dc_bus; service_access	bess_c_rate; bess_duration; bess_container_energy_density	none
submodule_allocation	major	ready	DC And AC Power Distribution / Earthing and lightning protection	Do the component candidates and interfaces assigned to Earthing and lightning protection make	Reviewer note confirming the submodule role, component allocation and any missing carriers or	power_distribution	protective_earth; service_access	bess_c_rate; bess_duration; bess_container_energy_density	none

Kind	Priority	Status	Scope	Reviewer Question	Evidence Required	Modules	Interfaces	Calculations	Blockers
submodule_allocation	major	ready	Safety Protection And Interlocks / Deflagration venting	engineering sense for its stated purpose?	capacity checks.				
				Do the component candidates and interfaces assigned to Deflagration venting make engineering sense for its stated purpose?	Reviewer note confirming the submodule role, component allocation and any missing carriers or capacity checks.	safety_protection	fire_boundary; alarm_bus; service_access	none	none

Showing first 70 of 81 engineering review questions; export CSV for the full pack.

Engineering Sanity Checks

Check	Status	Value	Expected	Interpretation
BESS C-Rate	pass	0.29 C	0.1C to 1.0C for typical grid-storage duration systems	Power-to-energy ratio implies 0.29C, equivalent to roughly 3.5 hours at rated power.
Rated Duration	pass	3.5 hours	1 to 6 hours for common containerised BESS applications	3.5 hours is plausible for a containerised grid-support BESS before detailed cell count and PCS sizing.
Container-Level Energy Density	pass	125 Wh/kg	50 to 200 Wh/kg at container/system level before detailed structural mass review	125 Wh/kg is within a plausible container-level range, but it still needs a mass roll-up by rack, PCS, HVAC and enclosure.

Engineering Calculation Ledger

3 calculations: 2 within deterministic envelope, 1 needing engineering review, 0 outside envelope, 0 blocked by missing inputs. These are arithmetic gates, not proof of real-world performance.

Calculation	Status	Formula	Result	Envelope	Interpretation	Evidence Required
BESS discharge duration	within envelope	capacity_mwh / power_mw	3.5 h	2 to 8 h is a common grid-storage concept envelope before duty-cycle proof.	3.5 h implied by nameplate capacity and PCS power.	Confirm use case, PCS rating, usable energy window, C-rate limits and warranty throughput.
BESS system energy density	within envelope	(capacity_mwh * 1,000,000) / mass_kg	125 Wh/kg	60 to 180 Wh/kg is a broad containerised LFP system envelope before packaging proof.	125 Wh/kg implied by energy and gross mass.	Review cell mass, racks, enclosure, HVAC, fire protection, transformer/PCS boundary and shipping mass allowance.
BESS annual nameplate throughput	needs review	capacity_mwh * 365	1277.5 MWh/year	Arithmetic only; real throughput depends on dispatch, degradation, auxiliary load and warranty.	1277.5 MWh/year assumes one full equivalent cycle per day.	Confirm cycles per year, usable depth of discharge, round-trip efficiency and degradation assumptions.

Engineering Assumption Ledger

21 assumptions: 3 brief-supported, 3 model-present, 7 needing engineering review, 8 needing source evidence, 0 blocked. Architecture blockers 0; BoM blockers 8.

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
Energy capacity	brief_requirement	brief supported	no	no	Energy capacity = 3.5 MWh is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Power rating	brief_requirement	brief supported	no	no	Power rating = 1 MW is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Mass constraint	brief_requirement	brief supported	no	no	Mass constraint = 28000 kg is treated as a design input, not as independently verified truth.	Confirm the requirement value, tolerance, duty cycle, environment and acceptance test with the project owner.
Annual MWh delivered	derived_metric	review required	no	no	Annual MWh delivered uses capacity_mwh * cycles_per_year * efficiency with medium confidence.	Review the formula, input values and operating envelope before treating this metric as a performance claim.
Estimated CAPEX	derived_metric	source required	no	yes	Estimated CAPEX uses an implicit model with low confidence.	Admit source-backed BoM pricing before treating this cost metric as a claim.
Estimated annual OPEX	derived_metric	source required	no	yes	Estimated annual OPEX uses an implicit model with low confidence.	Admit source-backed BoM pricing before treating this cost metric as a claim.
BESS C-Rate	sanity_envelope	review required	no	no	BESS C-Rate currently evaluates as pass against 0.1C to 1.0C for typical grid-storage duration systems.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
Rated Duration	sanity_envelope	review required	no	no	Rated Duration currently evaluates as	Supply an engineering calculation,

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					pass against 1 to 6 hours for common containerised BESS applications.	reviewer note, test plan or datasheet-backed rationale for this envelope.
Container-Level Energy Density	sanity_envelope	review required	no	no	Container-Level Energy Density currently evaluates as pass against 50 to 200 Wh/kg at container/system level before detailed structural mass review.	Supply an engineering calculation, reviewer note, test plan or datasheet-backed rationale for this envelope.
energy_storage_source -> energy_conversion_transduction	interface_closure	model present	no	no	dc_bus is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
energy_storage_source -> environmental_interface	interface_closure	model present	no	no	thermal_loop is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
energy_storage_source -> control_compute_communication	interface_closure	model present	no	no	bms_network is named on both endpoint modules in the generated architecture.	Review the interface carrier, direction, capacity, safety margin and failure mode before detailed design or sourcing.
LFP prismatic cells	critical_component	source required	no	yes	LFP prismatic cells is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
DC contactor or breaker	critical_component	source required	no	yes	DC contactor or breaker is a critical architecture-derived	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					component candidate with quantity 1 candidate.	and evidence reference through sourcing intake.
PCS inverter	critical_component	source required	no	yes	PCS inverter is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
BMS master controller	critical_component	source required	no	yes	BMS master controller is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Fire detection/suppression	critical_component	source required	no	yes	Fire detection/suppression is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
Thermal management loop	critical_component	source required	no	yes	Thermal management loop is a critical architecture-derived component candidate with quantity 1 candidate.	Admit supplier, manufacturer, MPN where applicable, unit cost, lead time and evidence reference through sourcing intake.
IEC-62619	compliance_evidence	review required	no	no	Industrial lithium battery safety is relevant to this product class in Global / UK-adopted.	Cell certificates and pack-level safety validation.
UL-9540A	compliance_evidence	review required	no	no	Thermal runaway fire propagation test is relevant to this product class in US / investor diligence.	Cell, module and unit-level propagation evidence.
G99	compliance_evidence	review required	no	no	UK grid connection	Protection settings, relay

Scope	Category	Status	Blocks Architecture	Blocks BoM	Assumption	Evidence Required
					requirements is relevant to this product class in UK.	test certificates and commissioning records.

Engineering Verification Plan

30 activities: 20 ready for review, 5 open, 5 blocked.

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Review Energy Storage Source architecture allocation	ready	design_review	Energy Storage Source	capacity_mwh	dc_bus; thermal_loop; bms_network; mechanical_mounts; bms_tap_harness; service_access; sensor_harness; hardwired_trip	lfp_prismatic_cells; cell_to_cell_busbar; cell_terminal_hardware_set; cell_insulation_pad; module_fuse; module_steel_frame; compression_plate; compression_tie_rod_set; plus 21 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Energy Storage Source requirement calculations	ready	calculation	Energy Storage Source	capacity_mwh	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Collect source-backed evidence for Energy Storage Source critical components	blocked	source_evidence	Energy Storage Source	capacity_mwh	none	lfp_prismatic_cells; dc_contactor_or_breaker	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved	lfp_prismatic_cells dc_contactor_or_breaker

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	
Review Power Conversion System architecture allocation	ready	design_review	Power Conversion System	capacity_mwh; power_mw	dc_bus; ac_bus; modbus_tcp; thermal_path; hardwired_trip; protective_earth	pcs_inverter; igbt_power_module; dc_link_capacitor_bank; gate_driver_board; inverter_control_board; liquid_cooled_heatsink; cast_resin_transformer; hv_winding; plus 12 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Power Conversion System requirement calculations	ready	calculation	Power Conversion System	capacity_mwh; power_mw	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Collect source-backed evidence for Power Conversion System critical components	blocked	source_evidence	Power Conversion System	capacity_mwh; power_mw	none	pcs_inverter	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	pcs_inverter
Review Container Structure And Fit-Out architecture allocation	ready	design_review	Container Structure And Fit-Out	mass_kg	mechanical_mounts; service_access; fire_boundary; lifting_points; protective_earth	corten_steel_side_panel; corner_casting; marine_plywood_floor; roof_panel; steel_stud_frame; fire_rated_board; mineral_wool_insulation; intumescent_sealant; plus 9 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Container Structure And Fit-Out requirement calculations	ready	calculation	Container Structure And Fit-Out	mass_kg	none	none	Each linked requirement has a calculation, sanity check or clearly	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	
Review Sensing And Instrumentation architecture allocation	ready	design_review	Sensing And Instrumentation	none	sensor_bus; alarm_bus; bms_network; site_scada; dc_bus; thermal_loop	dc_voltage_transducer; hall_current_sensor; insulation_monitoring_device; auxiliary_metering_transducer; temperature_humidity_sensor; hydrogen_detector; smoke_aspirating_detector; door_position_switch; plus 9 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Review Control Compute And Communication architecture allocation	ready	design_review	Control Compute And Communication	power_mw	bms_network; modbus_tcp; sensor_bus; alarm_bus; site_scada; service_access; hardwired_trip	bms_master_controller; isolation_interface; contactor_driver_output; ; soc_estimation_firmware; industrial_controller; i_o_expansion_module; network_switch; cellular_router; plus 9 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							issue is attached to the module.	
Verify Control Compute And Communication requirement calculations	ready	calculation	Control Compute And Communication	power_mw	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Collect source-backed evidence for Control Compute And Communication critical components	blocked	source_evidence	Control Compute And Communication	power_mw	none	bms_master_controller	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	bms_master_controller
Review Safety Protection And Interlocks architecture allocation	ready	design_review	Safety Protection And Interlocks	none	alarm_bus; hardwired_trip; fire_boundary; dc_bus; ac_bus; service_access;	fire_detection_suppression; clean_agent_cylinder; release_solenoid; nozzle_pipe_work;	Every submodule has a stated purpose and at least one component candidate. Interfaces	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
					protective_earth	pressure_switch; emergency_stop_mushroom_button; safety_relay; dc_shunt_trip_output; plus 10 more	named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	
Collect source-backed evidence for Safety Protection And Interlocks critical components	blocked	source_evidence	Safety Protection And Interlocks	none	none	fire_detection_suppression	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	fire_detection_suppression

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Review Environmental Interface architecture allocation	ready	design_review	Environmental Interface	none	thermal_loop; thermal_patching; electrical_power; sensor_bus; service_access; fire_boundary	container_hvac_unit; air_filter_element; condensate_drain_kit; hvac_controller; intake_louvre; exhaust_louvre; ec_fan_assembly; airflow_baffle; plus 9 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Review DC And AC Power Distribution architecture allocation	ready	design_review	DC And AC Power Distribution	capacity_mwh; power_mw	dc_bus; ac_bus; electrical_power; protective_earth; hardwired_trip; service_access	copper_dc_busbar; busbar_support_insulator; busbar_shroud; torque_witness_marker; moulded_case_circuit_breaker; ac_contactor; metering_ct_set; terminal_block_assembly; plus 9 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify DC And AC Power Distribution requirement calculations	ready	calculation	DC And AC Power Distribution	capacity_mwh; power_mw	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	
Review Maintenance And Serviceability architecture allocation	ready	design_review	Maintenance And Serviceability	mass_kg	service_access; lifting_points; protective_earth; sensor_bus; dc_bus; electrical_power; mechanical_mounts	lockable_access_door; door_seal_gasket; panic_release_latch; door_stay_arm; led_service_luminaire; emergency_light_fitting; maintenance_receptacle; lighting_switchgear; plus 9 more	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	none
Verify Maintenance And Serviceability requirement calculations	ready	calculation	Maintenance And Serviceability	mass_kg	none	none	Each linked requirement has a calculation, sanity check or clearly stated deferred validation path. No linked sanity check is failing. Warnings have named follow-up calculations or test evidence.	none
Review Coolant Circulation Process	ready	design_review	Coolant Circulation Process	none	thermal_loop; sensor_bus; service_acc	thermal_management_loop; coolant_circ	Every submodule has a stated purpose	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
architecture allocation					ess; electrical_power	ulation_pump; pump_vfd; check_valve; isolation_valve; coolant_filter_housing; replaceable_filter_cartridge; glycol_concentration_sensor; plus 10 more	and at least one component candidate. Interfaces named on the module are carried by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	
Collect source-backed evidence for Coolant Circulation Process critical components	blocked	source_evidence	Coolant Circulation Process	none	none	thermal_management_loop	Supplier name, manufacturer or equivalent, MPN where applicable, unit cost and source URL/reference are present. Evidence is explicitly retrieved and admissible under the sourcing admission protocol. Claimed quantity and component identity match the architecture candidate line.	thermal_management_loop
Review Operator HMI And Labelling architecture allocation	ready	design_review	Operator HMI And Labelling	none	site_scada; service_access; alarm_bus	industrial_touchscreen; panel_pc; key_switch; alarm_buzzer_beacon; high_voltage_warning_label; arc_flash_boundary_label;	Every submodule has a stated purpose and at least one component candidate. Interfaces named on the module are carried	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
						fire_suppression_instruction_placard; asset_qr_label; plus 9 more	by the responsible submodule where the design depends on that interface. No blocker or major architecture issue is attached to the module.	
Verify dc_bus interface contract	ready	interface_review	Energy Storage Source -> Power Conversion System	none	dc_bus	lfp_prismatic_cells; cell_to_cell_busbar; cell_terminal_hardware_set; cell_insulation_pad; module_fuse; dc_contactor_or_breaker; precharge_resistor; hv_fuse_holder; plus 8 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Verify thermal_loop interface contract	ready	interface_review	Energy Storage Source -> Environmental Interface	none	thermal_loop	lfp_prismatic_cells; cell_to_cell_busbar; cell_terminal_hardware_set; cell_insulation_pad; module_fuse; rack_cold_plate_manifold; thermally_conductive_gap_pad; coolant_quick_connect_or_pair; plus 15 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
Verify bms_network interface contract	ready	interface_review	Energy Storage Source -> Control Compute And Communication	none	bms_network	bms_slave_pcb; afe_monitor_ic; ntc_thermistors_harness; isolated_cann_transceiver; conformal_coating; bms_master_controller; isolation_interface; contactor_driver_output; plus 1 more	Both endpoint modules declare the interface. At least one submodule carrier is named on each endpoint. Electrical, thermal, fluid, mechanical or data capacity is validated by the next detailed design pass.	none
Plan evidence for IEC-62619	open	compliance_review	System Compliance	none	none	none	Cell certificates and pack-level safety validation.	Compliance evidence not yet collected in scratch prototype.
Plan evidence for UL-9540A	open	compliance_review	System Compliance	none	none	none	Cell, module and unit-level propagation evidence.	Compliance evidence not yet collected in scratch prototype.
Plan evidence for G99	open	compliance_review	System Compliance	none	none	none	Protection settings, relay test certificates and commissioning records.	Compliance evidence not yet collected in scratch prototype.
Review risk control for Thermal runaway propagation	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an	Risk evidence not yet reviewed in scratch prototype.

Activity	Status	Evidence	Scope	Requirements	Interfaces	Components	Acceptance Criteria	Blockers
							owner before publication.	
Review risk control for DC arc fault	open	compliance_review	System Risk Register	none	none	none	Severity, occurrence and detection ratings have been reviewed by a domain owner. Mitigation is accepted, corrected or converted into a design action. Any residual high-risk item has an owner before publication.	Risk evidence not yet reviewed in scratch prototype.

Verification Evidence Ledger

0 accepted, 0 rejected, 0 deferred, 25 pending, 0 blocked without evidence. Review coverage 0; acceptance ratio 0. Source-evidence activities remain in sourcing intake.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Review Energy Storage Source architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Energy Storage Source requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Energy Storage Source critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Power Conversion System architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Power Conversion System requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Power Conversion System critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Container Structure And Fit-Out architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Container Structure And Fit-Out requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Review Sensing And Instrumentation architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Review Control Compute And Communication architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Control Compute And Communication requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Control Compute And Communication critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Review Safety Protection And Interlocks architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Safety Protection And Interlocks critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Environmental Interface architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Review DC And AC Power Distribution architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify DC And AC Power Distribution requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Review Maintenance And Serviceability architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify Maintenance And Serviceability requirement calculations	pending	calculation			Collect reviewer evidence through the verification intake template.
Review Coolant Circulation Process architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Collect source-backed evidence for Coolant Circulation Process critical components	source required	source_evidence			Use sourcing intake for supplier, manufacturer, MPN, cost and lead-time evidence.
Review Operator HMI And Labelling architecture allocation	pending	design_review			Collect reviewer evidence through the verification intake template.
Verify dc_bus interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Verify thermal_loop interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.

Activity	Ledger Status	Evidence Kind	Reviewer	Evidence Ref	Residual Action
Verify bms_network interface contract	pending	interface_review			Collect reviewer evidence through the verification intake template.
Plan evidence for IEC-62619	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Plan evidence for UL-9540A	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Plan evidence for G99	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Review risk control for Thermal runaway propagation	pending	compliance_review			Collect reviewer evidence through the verification intake template.
Review risk control for DC arc fault	pending	compliance_review			Collect reviewer evidence through the verification intake template.

Energy Storage Source

Stores roughly 3.5 MWh usable energy in rack-mounted LFP cell assemblies.

- dc_bus
- thermal_loop
- bms_network
- mechanical_mounts

Cell string

Electrochemical storage path assembled from series/parallel LFP cells.

Components: LFP prismatic cells, cell-to-cell busbar, cell terminal hardware set, cell insulation pad, module fuse

Rack structure

Constrains cell modules during transport, vibration and cycling expansion.

Components: module steel frame, compression plate, compression tie rod set, module top cover, rack label plate

BMS slave board

Measures cell voltage and temperature at rack level.

Components: BMS slave PCB, AFE monitor IC, NTC thermistor harness, isolated CAN transceiver, conformal coating

DC string output

Combines rack outputs into a protected high-voltage DC path.

Components: DC contactor or breaker, precharge resistor, HV fuse holder, pack current sensor, service disconnect

Module interconnect harness

Routes low-voltage sensing, communications and temperature signals across rack assemblies.

Components: cell voltage sense loom, rack CAN drop cable, harness strain relief rail, low-smoke cable sleeve

Rack thermal interface

Couples cell modules to the container thermal loop while preserving service replacement access.

Components: rack cold plate manifold, thermally conductive gap pad, coolant quick connector pair, rack drip tray, leak detection rope

Power Conversion System

Converts the battery DC bus into 1 MW grid-compatible AC output.

- dc_bus
- ac_bus
- modbus_tcp
- thermal_path
- hardwired_trip
- protective_earth

PCS inverter bridge

Performs bidirectional DC/AC conversion.

Components: PCS inverter, IGBT power module, DC link capacitor bank, gate driver board, inverter control board, liquid-cooled heatsink

Transformer stage

Adapts PCS output voltage to site connection voltage.

Components: cast resin transformer, HV winding, LV winding, temperature probe, anti-vibration mount

Grid filter

Controls harmonics and conducted emissions.

Components: grid filter choke, AC filter capacitor, EMI filter assembly, protective earth bar

Grid protection relay

Supervises grid voltage, frequency and trip settings at the point of connection.

Components: grid protection relay, voltage sensing fuse set, relay test block, trip circuit monitor, G99 settings label

Container Structure And Fit-Out

Provides weatherproof containment, transport structure and internal segregation.

- mechanical_mounts
- service_access
- fire_boundary
- lifting_points
- protective_earth

ISO container shell

Forms the weather-tight transportable enclosure.

Components: corten steel side panel, corner casting, marine plywood floor, roof panel

Internal partitions

Separates battery, PCS, controls and service zones.

Components: steel stud frame, fire-rated board, mineral wool insulation, intumescent sealant

Rack mounting rails

Anchors battery racks against transport and seismic loads.

Components: floor reinforcement plate, rack mounting rail, vibration isolator, anchor bolt set

Cable penetration sealing

Seals power, control and auxiliary cable entries through container fire and weather boundaries.

Components: gland plate assembly, firestop collar set, EMC cable gland, rain drip shield, penetration schedule label

Sensing And Instrumentation

Measures electrical, thermal and environmental state for operation and safety.

- sensor_bus
- alarm_bus
- bms_network
- site_scada

Electrical measurement

Measures voltage, current and insulation state.

Components: DC voltage transducer, Hall current sensor, insulation monitoring device, auxiliary metering transducer

Environmental sensors

Measures container air conditions and gas hazards.

Components: temperature humidity sensor, hydrogen detector, smoke aspirating detector, door position switch

Thermal sensors

Tracks coolant and heat exchanger temperatures.

Components: coolant inlet thermistor, coolant outlet thermistor, flow meter, pressure transducer

Event recording

Captures timestamped trips, alarms and measurements for engineering review.

Components: fault recorder module, event timestamp logger, digital input card, local historian buffer, calibration certificate holder

Control Compute And Communication

Coordinates BMS, PCS, auxiliaries and site-level dispatch.

- bms_network
- modbus_tcp
- sensor_bus
- alarm_bus
- site_scada
- service_access

BMS master controller

Supervises pack state and protective limits.

Components: BMS master controller, isolation interface, contactor driver output, SOC estimation firmware

EMS controller

Dispatches power and monitors system state.

Components: industrial controller, I/O expansion module, network switch, cellular router

Communications gateway

Bridges local controls to site and remote systems.

Components: Modbus TCP gateway, SCADA protocol adapter, time synchronisation module, cybersecurity firewall

Cybersecurity monitoring

Manages secure access, logging and recovery data for remote operation.

Components: VPN access appliance, syslog collector, account key switch, firmware escrow tag, configuration backup module

Safety Protection And Interlocks

Detects unsafe conditions and forces the system into a protected state.

- alarm_bus
- hardwired_trip
- fire_boundary
- dc_bus
- ac_bus
- service_access

Fire suppression system

Detects and suppresses incipient battery-container fire.

Components: Fire detection/suppression, clean-agent cylinder, release solenoid, nozzle pipework, pressure switch

Emergency stop chain

Trips high-energy paths through hardwired safety logic.

Components: E-stop mushroom button, safety relay, DC shunt trip output, AC breaker trip coil

Surge and fault protection

Limits transient and overcurrent damage.

Components: DC surge protection device, AC surge protection device, arc flash label set, protective earth bonding kit

Deflagration venting

Provides a controlled pressure relief path if battery off-gas ignition occurs.

Components: pressure relief vent panel, vent microswitch, flame arrestor mesh, vent weather hood, post-event inspection tag

Environmental Interface

Maintains the container operating envelope across weather and heat loads.

- thermal_loop
- thermal_path
- electrical_power
- sensor_bus
- service_access

HVAC unit

Controls air temperature and humidity inside compartments.

Components: container HVAC unit, air filter element, condensate drain kit, HVAC controller

Ventilation path

Directs airflow around PCS, controls and service areas.

Components: intake louver, exhaust louver, EC fan assembly, airflow baffle

Thermal insulation

Reduces ambient heat transfer into the container.

Components: insulation panel, vapour barrier film, thermal break strip, seal gasket set

Heat rejection coil

Rejects battery and PCS heat from the coolant loop to outdoor air.

Components: dry cooler coil, EC condenser fan, fan contactor, coil guard grille, drain-safe condensate pan

DC And AC Power Distribution

Routes high-voltage DC, grid AC and auxiliary power through protected paths.

- dc_bus
- ac_bus
- electrical_power
- protective_earth
- hardwired_trip

Main DC busbars

Carries pack current from racks to PCS.

Components: copper DC busbar, busbar support insulator, busbar shroud, torque witness marker

AC switchgear

Connects PCS output to site AC connection.

Components: moulded-case circuit breaker, AC contactor, metering CT set, terminal block assembly

Auxiliary power distribution

Supplies controls, HVAC, pumps and service loads.

Components: auxiliary transformer, 24 V DC power supply, miniature circuit breaker bank, UPS module

Earthing and lightning protection

Bonds exposed metalwork and surge devices to a verifiable protective-earth network.

Components: main earth bar, earth stud array, lightning surge counter, equipotential bonding strap, earth continuity test point

Maintenance And Serviceability

Lets technicians inspect, isolate, lift and replace internal assemblies safely.

- service_access
- lifting_points
- protective_earth
- sensor_bus
- dc_bus

Access doors and hardware

Provides controlled entry to service zones.

Components: lockable access door, door seal gasket, panic release latch, door stay arm

Service lighting

Illuminates work zones during inspection.

Components: LED service luminaire, emergency light fitting, maintenance receptacle, lighting switchgear

Lifting and handling points

Supports transport, installation and module replacement.

Components: certified lifting lug, forklift pocket reinforcement, removable rack guide, service trolley interface

Commissioning ports

Provides controlled access points for electrical, coolant and controller commissioning checks.

Components: insulated test socket, coolant sampling port, laptop service port, lockout hasp rail, commissioning torque log plate

Coolant Circulation Process

Moves coolant through battery racks, heat exchangers and serviceable filters.

- thermal_loop
- sensor_bus
- service_access
- electrical_power

Coolant pump station

Circulates coolant through rack cold plates.

Components: Thermal management loop, coolant circulation pump, pump VFD, check valve, isolation valve

Filtration and treatment

Maintains coolant cleanliness and chemistry.

Components: coolant filter housing, replaceable filter cartridge, glycol concentration sensor, fill and drain manifold

Expansion and air removal

Accommodates fluid expansion and removes trapped air.

Components: expansion vessel, automatic air vent, pressure relief valve, low pressure switch

Heat exchanger manifold

Transfers rack heat into the outdoor heat rejection path while preserving service bypass options.

Components: plate heat exchanger, temperature balancing valve, purge valve set, service bypass loop, insulated hose set

Operator HMI And Labelling

Presents system status, controls access and communicates hazards to operators.

- site_scada
- service_access
- alarm_bus

Local HMI panel

Provides local status, alarms and manual controls.

Components: industrial touchscreen, panel PC, key switch, alarm buzzer beacon

External labelling

Communicates hazards and service instructions.

Components: high voltage warning label, arc flash boundary label, fire suppression instruction placard, asset QR label

Operator workflow aids

Supports safe inspection and reset actions.

Components: laminated isolation checklist, inspection log holder, status stack light, manual reset station

Remote notification interface

Routes actionable alarms to the site operator while preventing unsafe remote reset.

Components: operator notification relay, SMS alarm output module, alarm acknowledgement button, escalation contact label, remote reset inhibit relay

Bill of Materials

Pricing status: architecture-derived candidate BoM only. Costs, suppliers, manufacturers and part numbers have not been independently sourced.

Total CAPEX BoM: £0

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
LFP prismatic cells	1	candidate	unpriced	unpriced				
cell-to-cell busbar	1	candidate	unpriced	unpriced				
cell terminal hardware set	1	candidate	unpriced	unpriced				
cell insulation pad	1	candidate	unpriced	unpriced				
module fuse	1	candidate	unpriced	unpriced				
module steel frame	1	candidate	unpriced	unpriced				
compression plate	1	candidate	unpriced	unpriced				
compression tie rod set	1	candidate	unpriced	unpriced				
module top cover	1	candidate	unpriced	unpriced				
rack label plate	1	candidate	unpriced	unpriced				
BMS slave PCB	1	candidate	unpriced	unpriced				
AFE monitor IC	1	candidate	unpriced	unpriced				
NTC thermistor harness	1	candidate	unpriced	unpriced				
isolated CAN transceiver	1	candidate	unpriced	unpriced				
conformal coating	1	candidate	unpriced	unpriced				
DC contactor or breaker	1	candidate	unpriced	unpriced				
precharge resistor	1	candidate	unpriced	unpriced				
HV fuse holder	1	candidate	unpriced	unpriced				
pack current sensor	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
service disconnect	1	candidate	unpriced	unpriced				
cell voltage sense loom	1	candidate	unpriced	unpriced				
rack CAN drop cable	1	candidate	unpriced	unpriced				
harness strain relief rail	1	candidate	unpriced	unpriced				
low-smoke cable sleeve	1	candidate	unpriced	unpriced				
rack cold plate manifold	1	candidate	unpriced	unpriced				
thermally conductive gap pad	1	candidate	unpriced	unpriced				
coolant quick connector pair	1	candidate	unpriced	unpriced				
rack drip tray	1	candidate	unpriced	unpriced				
leak detection rope	1	candidate	unpriced	unpriced				
PCS inverter	1	candidate	unpriced	unpriced				
IGBT power module	1	candidate	unpriced	unpriced				
DC link capacitor bank	1	candidate	unpriced	unpriced				
gate driver board	1	candidate	unpriced	unpriced				
inverter control board	1	candidate	unpriced	unpriced				
liquid-cooled heatsink	1	candidate	unpriced	unpriced				
cast resin transformer	1	candidate	unpriced	unpriced				
HV winding	1	candidate	unpriced	unpriced				
LV winding	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
temperature probe	1	candidate	unpriced	unpriced				
anti-vibration mount	1	candidate	unpriced	unpriced				
grid filter choke	1	candidate	unpriced	unpriced				
AC filter capacitor	1	candidate	unpriced	unpriced				
EMI filter assembly	1	candidate	unpriced	unpriced				
protective earth bar	1	candidate	unpriced	unpriced				
grid protection relay	1	candidate	unpriced	unpriced				
voltage sensing fuse set	1	candidate	unpriced	unpriced				
relay test block	1	candidate	unpriced	unpriced				
trip circuit monitor	1	candidate	unpriced	unpriced				
G99 settings label	1	candidate	unpriced	unpriced				
corten steel side panel	1	candidate	unpriced	unpriced				
corner casting	1	candidate	unpriced	unpriced				
marine plywood floor	1	candidate	unpriced	unpriced				
roof panel	1	candidate	unpriced	unpriced				
steel stud frame	1	candidate	unpriced	unpriced				
fire-rated board	1	candidate	unpriced	unpriced				
mineral wool insulation	1	candidate	unpriced	unpriced				
intumescent sealant	1	candidate	unpriced	unpriced				
floor reinforcement	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
nt plate								
rack mounting rail	1	candidate	unpriced	unpriced				
vibration isolator	1	candidate	unpriced	unpriced				
anchor bolt set	1	candidate	unpriced	unpriced				
gland plate assembly	1	candidate	unpriced	unpriced				
firestop collar set	1	candidate	unpriced	unpriced				
EMC cable gland	1	candidate	unpriced	unpriced				
rain drip shield	1	candidate	unpriced	unpriced				
penetration schedule label	1	candidate	unpriced	unpriced				
DC voltage transducer	1	candidate	unpriced	unpriced				
Hall current sensor	1	candidate	unpriced	unpriced				
insulation monitoring device	1	candidate	unpriced	unpriced				
auxiliary metering transducer	1	candidate	unpriced	unpriced				
temperature humidity sensor	1	candidate	unpriced	unpriced				
hydrogen detector	1	candidate	unpriced	unpriced				
smoke aspirating detector	1	candidate	unpriced	unpriced				
door position switch	1	candidate	unpriced	unpriced				
coolant inlet thermistor	1	candidate	unpriced	unpriced				
coolant outlet thermistor	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
flow meter	1	candidate	unpriced	unpriced				
pressure transducer	1	candidate	unpriced	unpriced				
fault recorder module	1	candidate	unpriced	unpriced				
event timestamp logger	1	candidate	unpriced	unpriced				
digital input card	1	candidate	unpriced	unpriced				
local historian buffer	1	candidate	unpriced	unpriced				
calibration certificate holder	1	candidate	unpriced	unpriced				
BMS master controller	1	candidate	unpriced	unpriced				
isolation interface	1	candidate	unpriced	unpriced				
contactor driver output	1	candidate	unpriced	unpriced				
SOC estimation firmware	1	candidate	unpriced	unpriced				
industrial controller	1	candidate	unpriced	unpriced				
I/O expansion module	1	candidate	unpriced	unpriced				
network switch	1	candidate	unpriced	unpriced				
cellular router	1	candidate	unpriced	unpriced				
Modbus TCP gateway	1	candidate	unpriced	unpriced				
SCADA protocol adapter	1	candidate	unpriced	unpriced				
time synchronisation module	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
cybersecurity firewall	1	candidate	unpriced	unpriced				
VPN access appliance	1	candidate	unpriced	unpriced				
syslog collector	1	candidate	unpriced	unpriced				
account key switch	1	candidate	unpriced	unpriced				
firmware escrow tag	1	candidate	unpriced	unpriced				
configuration backup module	1	candidate	unpriced	unpriced				
Fire detection/suppression	1	candidate	unpriced	unpriced				
clean-agent cylinder	1	candidate	unpriced	unpriced				
release solenoid	1	candidate	unpriced	unpriced				
nozzle pipework	1	candidate	unpriced	unpriced				
pressure switch	1	candidate	unpriced	unpriced				
E-stop mushroom button	1	candidate	unpriced	unpriced				
safety relay	1	candidate	unpriced	unpriced				
DC shunt trip output	1	candidate	unpriced	unpriced				
AC breaker trip coil	1	candidate	unpriced	unpriced				
DC surge protection device	1	candidate	unpriced	unpriced				
AC surge protection device	1	candidate	unpriced	unpriced				
arc flash label set	1	candidate	unpriced	unpriced				
protective earth bonding kit	1	candidate	unpriced	unpriced				
pressure relief vent	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
panel								
vent microswitch	1	candidate	unpriced	unpriced				
flame arrestor mesh	1	candidate	unpriced	unpriced				
vent weather hood	1	candidate	unpriced	unpriced				
post-event inspection tag	1	candidate	unpriced	unpriced				
container HVAC unit	1	candidate	unpriced	unpriced				
air filter element	1	candidate	unpriced	unpriced				
condensate drain kit	1	candidate	unpriced	unpriced				
HVAC controller	1	candidate	unpriced	unpriced				
intake louver	1	candidate	unpriced	unpriced				
exhaust louver	1	candidate	unpriced	unpriced				
EC fan assembly	1	candidate	unpriced	unpriced				
airflow baffle	1	candidate	unpriced	unpriced				
insulation panel	1	candidate	unpriced	unpriced				
vapour barrier film	1	candidate	unpriced	unpriced				
thermal break strip	1	candidate	unpriced	unpriced				
seal gasket set	1	candidate	unpriced	unpriced				
dry cooler coil	1	candidate	unpriced	unpriced				
EC condenser fan	1	candidate	unpriced	unpriced				
fan contactor	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
coil guard grille	1	candidate	unpriced	unpriced				
drain-safe condensate pan	1	candidate	unpriced	unpriced				
copper DC busbar	1	candidate	unpriced	unpriced				
busbar support insulator	1	candidate	unpriced	unpriced				
busbar shroud	1	candidate	unpriced	unpriced				
torque witness marker	1	candidate	unpriced	unpriced				
moulded-case circuit breaker	1	candidate	unpriced	unpriced				
AC contactor	1	candidate	unpriced	unpriced				
metering CT set	1	candidate	unpriced	unpriced				
terminal block assembly	1	candidate	unpriced	unpriced				
auxiliary transformer	1	candidate	unpriced	unpriced				
24 V DC power supply	1	candidate	unpriced	unpriced				
miniature circuit breaker bank	1	candidate	unpriced	unpriced				
UPS module	1	candidate	unpriced	unpriced				
main earth bar	1	candidate	unpriced	unpriced				
earth stud array	1	candidate	unpriced	unpriced				
lightning surge counter	1	candidate	unpriced	unpriced				
equipotential bonding strap	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
earth continuity test point	1	candidate	unpriced	unpriced				
lockable access door	1	candidate	unpriced	unpriced				
door seal gasket	1	candidate	unpriced	unpriced				
panic release latch	1	candidate	unpriced	unpriced				
door stay arm	1	candidate	unpriced	unpriced				
LED service luminaire	1	candidate	unpriced	unpriced				
emergency light fitting	1	candidate	unpriced	unpriced				
maintenance receptacle	1	candidate	unpriced	unpriced				
lighting switchgear	1	candidate	unpriced	unpriced				
certified lifting lug	1	candidate	unpriced	unpriced				
forklift pocket reinforcement	1	candidate	unpriced	unpriced				
removable rack guide	1	candidate	unpriced	unpriced				
service trolley interface	1	candidate	unpriced	unpriced				
insulated test socket	1	candidate	unpriced	unpriced				
coolant sampling port	1	candidate	unpriced	unpriced				
laptop service port	1	candidate	unpriced	unpriced				
lockout hasp rail	1	candidate	unpriced	unpriced				
commissioning torque log plate	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
Thermal management loop	1	candidate	unpriced	unpriced				
coolant circulation pump	1	candidate	unpriced	unpriced				
pump VFD	1	candidate	unpriced	unpriced				
check valve	1	candidate	unpriced	unpriced				
isolation valve	1	candidate	unpriced	unpriced				
coolant filter housing	1	candidate	unpriced	unpriced				
replaceable filter cartridge	1	candidate	unpriced	unpriced				
glycol concentration sensor	1	candidate	unpriced	unpriced				
fill and drain manifold	1	candidate	unpriced	unpriced				
expansion vessel	1	candidate	unpriced	unpriced				
automatic air vent	1	candidate	unpriced	unpriced				
pressure relief valve	1	candidate	unpriced	unpriced				
low pressure switch	1	candidate	unpriced	unpriced				
plate heat exchanger	1	candidate	unpriced	unpriced				
temperature balancing valve	1	candidate	unpriced	unpriced				
purge valve set	1	candidate	unpriced	unpriced				
service bypass loop	1	candidate	unpriced	unpriced				
insulated hose set	1	candidate	unpriced	unpriced				
industrial touchscreen	1	candidate	unpriced	unpriced				
panel PC	1	candidate	unpriced	unpriced				

Line	Qty	Unit	Unit Cost	Total	Supplier	Manufacturer	MPN	Lead Time
key switch	1	candidate	unpriced	unpriced				
alarm buzzer beacon	1	candidate	unpriced	unpriced				
high voltage warning label	1	candidate	unpriced	unpriced				
arc flash boundary label	1	candidate	unpriced	unpriced				
fire suppression instruction placard	1	candidate	unpriced	unpriced				
asset QR label	1	candidate	unpriced	unpriced				
laminated isolation checklist	1	candidate	unpriced	unpriced				
inspection log holder	1	candidate	unpriced	unpriced				
status stack light	1	candidate	unpriced	unpriced				
manual reset station	1	candidate	unpriced	unpriced				
operator notification relay	1	candidate	unpriced	unpriced				
SMS alarm output module	1	candidate	unpriced	unpriced				
alarm acknowledg ement button	1	candidate	unpriced	unpriced				
escalation contact label	1	candidate	unpriced	unpriced				
remote reset inhibit relay	1	candidate	unpriced	unpriced				

Cost And Sourcing

Sourcing Admission

Status	Candidate Lines	Admitted Prices	Unpriced Lines	Unpriced Critical Lines
not_started	204	0	204	6

Critical Sourcing Worklist

Component	Qty	Reason
LFP prismatic cells	1 candidate	Required class-critical line must be sourced before BoM can pass.
DC contactor or breaker	1 candidate	Required class-critical line must be sourced before BoM can pass.
PCS inverter	1 candidate	Required class-critical line must be sourced before BoM can pass.
BMS master controller	1 candidate	Required class-critical line must be sourced before BoM can pass.
Fire detection/suppression	1 candidate	Required class-critical line must be sourced before BoM can pass.
Thermal management loop	1 candidate	Required class-critical line must be sourced before BoM can pass.

No rejected source records.

BoM Sourcing Ledger

0/204 priced lines admitted. Critical coverage 0/6. Rejected evidence records: 0. Cost/manufacturer/MPN fields stay blank until evidence is admitted.

Component identity check: no duplicate componentWorldId groups detected.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
LFP prismatic cells	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacture r/MPN/unit -cost evidence before BoM can pass.
cell-to-cell busbar	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
cell terminal hardware set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
cell insulation pad	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
module fuse	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
module steel frame	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									planning starts.
compression plate	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
compression tie rod set	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
module top cover	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
rack label plate	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
BMS slave PCB	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
AFE monitor IC	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									procurement planning starts.
NTC thermistor harness	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
isolated CAN transceiver	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
conformal coating	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
DC contactor or breaker	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
precharge resistor	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
HV fuse holder	candidate unpriced	unique	candidate	unpriced					Source after critical lines or

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									when detailed procurement planning starts.
pack current sensor	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
service disconnect	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
cell voltage sense loom	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
rack CAN drop cable	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
harness strain relief rail	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
low-smoke cable sleeve	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
rack cold plate manifold	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
thermally conductive gap pad	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
coolant quick connector pair	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
rack drip tray	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
leak detection rope	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									planning starts.
PCS inverter	critical unpriced	unique	critical	unpriced					Collect admissible supplier/m anufacturer/MPN/unit -cost evidence before BoM can pass.
IGBT power module	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
DC link capacitor bank	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
gate driver board	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
inverter control board	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
liquid-cooled heatsink	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed

Line	Status	Duplicate	Priority	Unit Cost	Supplier	Manufacturer	MPN	Evidence Ref	Next Action
									procurement planning starts.
cast resin transformer	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
HV winding	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
LV winding	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
temperature probe	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.
anti-vibration mount	candidate unpriced	unique	candidate	unpriced					Source after critical lines or when detailed procurement planning starts.

Showing first 40 of 204 ledger rows; export CSV for the full line-by-line ledger.

0 source-backed claims, 24 missing-source claims, 0 provenance violations. Critical missing-source claims: 24. Supplier, manufacturer, MPN and cost fields must appear here with source refs before procurement use.

Line	Claim Field	Status	Value	Source Ref	Next Action
LFP prismatic cells	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
LFP prismatic cells	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
LFP prismatic cells	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
LFP prismatic cells	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
DC contactor or breaker	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
DC contactor or breaker	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
DC contactor or breaker	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
DC contactor or breaker	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
PCS inverter	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
PCS inverter	manufacturer	missing source			Admit source-backed

Line	Claim Field	Status	Value	Source Ref	Next Action
					supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
PCS inverter	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
PCS inverter	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
BMS master controller	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
BMS master controller	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
BMS master controller	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
BMS master controller	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Fire detection/suppression	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Fire detection/suppression	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Fire detection/suppression	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost

Line	Claim Field	Status	Value	Source Ref	Next Action
					evidence before BoM can pass.
Fire detection/suppression	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Thermal management loop	supplier	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Thermal management loop	manufacturer	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Thermal management loop	mpn	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.
Thermal management loop	unit_cost_gbp	missing source			Admit source-backed supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.

Source Reference Quality Gate

No sourcing evidence references have been admitted. Manufacturer, MPN and cost fields must remain blank until source-backed evidence is admitted through sourcing intake.

BoM Evidence Trace Matrix

0/204 BoM row(s) are production-eligible; 0 protocol-only, 0 source-reference blocked, 6 critical unsourced. Priced-review rows 0; procurement use no; next row bom-trace:energy_storage_source_cell_string_lfp_prismatic_cells.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
LFP prismatic cells	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
cell-to-cell busbar	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
cell terminal hardware set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
cell insulation pad	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
module fuse	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
module steel frame	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									or procurement planning starts.
compression plate	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
compression tie rod set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
module top cover	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
rack label plate	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
BMS slave PCB	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									planning starts.
AFE monitor IC	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
NTC thermistor harness	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
isolated CAN transceiver	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
conformal coating	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
DC contactor or breaker	critical unsource	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
precharge resistor	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									critical sourcing is complete or procurement planning starts.
HV fuse holder	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
pack current sensor	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
service disconnect	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
cell voltage sense loom	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
rack CAN drop cable	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									procurement planning starts.
harness strain relief rail	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
low-smoke cable sleeve	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
rack cold plate manifold	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
thermally conductive gap pad	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
coolant quick connector pair	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
rack drip tray	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
leak detection rope	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
PCS inverter	critical unsourced	critical	unpriced			0/4	no	no	Collect admissible source evidence for this critical line before BoM can pass.
IGBT power module	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
DC link capacitor bank	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
gate driver board	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									or procurement planning starts.
inverter control board	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
liquid-cooled heatsink	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
cast resin transformer	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
HV winding	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
LV winding	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									planning starts.
temperature probe	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
anti-vibration mount	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
grid filter choke	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
AC filter capacitor	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
EMI filter assembly	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
protective earth bar	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
grid protection relay	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
voltage sensing fuse set	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
relay test block	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
trip circuit monitor	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
G99 settings label	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical

Line	Trace Status	Priority	Unit Cost	Source Status	Source Class	Required Claims	Priced Review	Procurement	Required Action
									sourcing is complete or procurement planning starts.
corten steel side panel	candidate only	candidate	unpriced			0/4	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Showing first 50 of 204 BoM trace rows; export CSV for the full trace.

BoM Evidence Closure Plan

204 closure row(s): 6 ready, 0 blocked, 198 deferred. Actions: 6 collect source, 0 repair, 0 replace protocol, 198 candidate deferrals. Procurement blockers 6; priced-review blockers 6; next row bom-close:control_compute_communication_bms_master_bms_master_controller.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
1	BMS master controller	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
2	PCS inverter	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
3	LFP prismatic cells	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
4	DC contactor or breaker	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
5	Thermal management loop	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.
6	Fire detection/suppression	ready	blocker	collect source evidence	critical unsourced	supplierName; manufacturer; mpn; unitCostGBP; evidence.ref	yes	yes	Collect admissible source evidence for this critical line before BoM can pass.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
7	contactor driver output	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
8	isolation interface	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
9	SOC estimation firmware	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
10	cybersecurity firewall	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
11	Modbus TCP gateway	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
12	SCADA protocol adapter	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
13	time synchronisation module	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
14	account key switch	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
15	configuration backup module	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
16	firmware escrow tag	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
17	syslog collector	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
18	VPN access appliance	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
19	cellular router	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
20	I/O expansion module	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
21	industrial controller	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
22	network switch	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
23	AC filter capacitor	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
24	EMI filter assembly	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
25	grid filter choke	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
26	protective earth bar	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
27	G99 settings label	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
28	grid protection relay	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
29	relay test block	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
30	trip circuit monitor	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
31	voltage sensing fuse set	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
32	DC link capacitor bank	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
33	gate driver board	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
34	IGBT power module	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
35	inverter control board	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
36	liquid-cooled heatsink	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
37	anti-vibration mount	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
38	cast resin transformer	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
39	HV winding	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
40	LV winding	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
41	temperature probe	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
42	AFE monitor IC	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
43	BMS slave PCB	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
44	conformal coating	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
45	isolated CAN transceiver	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
46	NTC thermistor harness	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

#	Line	Status	Priority	Action	Trace Status	Required Evidence	Blocks Priced Review	Blocks Procurement	Required Action
47	cell insulation pad	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
48	cell terminal hardware set	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
49	cell-to-cell busbar	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.
50	module fuse	deferr ed	minor	defer candidate sourcing	candidate only	no immediate evidence required unless this candidate is promoted to procurement scope	no	no	Keep as candidate-only until critical sourcing is complete or procurement planning starts.

Showing first 50 of 204 BoM closure rows; export CSV for the full queue.

Sourcing Batch Plan

2 batch(es): 1 active, 0 waiting, 1 deferred. Active rows 6; deferred rows 198. Critical source rows 6; repair rows 0; protocol replacements 0. Next batch sourcing-batch:2:critical_source_collection; next item critical_source_collection:1:bms_master_controller.

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
2	Collect Critical Source Evidence	active	blocker	BMS master controller	BMS master controller; BMS master controller manufacturer part number; BMS master controller catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	PCS inverter	PCS inverter; PCS inverter manufacturer part number; PCS inverter catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	LFP prismatic cells	LFP prismatic cells; LFP prismatic cells manufacturer part number; LFP prismatic cells catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	DC contactor or breaker	DC contactor or breaker; DC contactor or breaker manufacturer part number; DC contactor or breaker catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
2	Collect Critical Source Evidence	active	blocker	Thermal management loop	Thermal management loop; Thermal management loop manufacturer part number; Thermal management loop catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
2	Collect Critical Source Evidence	active	blocker	Fire detection/suppression	Fire detection/suppression; Fire detection/suppression manufacturer part number; Fire detection/suppression catalogue price GBP	supplierName; manufacturer; mpn; unitCostGbp; evidence.ref	yes	source evidence admitted
6	Defer Candidate-Only Sourcing	deferred	minor	contactor driver output	contactor driver output; contactor driver output manufacturer part number; contactor driver output catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	isolation interface	isolation interface; isolation interface manufacturer part number; isolation interface catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	SOC estimation firmware	SOC estimation firmware; SOC estimation firmware	no immediate evidence required unless this candidate is	no	candidate remains out of procurement totals

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
					manufacturer part number; SOC estimation firmware catalogue price GBP	promoted to procurement scope		
6	Defer Candidate-Only Sourcing	deferred	minor	cybersecurity firewall	cybersecurity firewall; cybersecurity firewall manufacturer part number; cybersecurity firewall catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	Modbus TCP gateway	Modbus TCP gateway; Modbus TCP gateway manufacturer part number; Modbus TCP gateway catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	SCADA protocol adapter	SCADA protocol adapter; SCADA protocol adapter manufacturer part number; SCADA protocol adapter catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	time synchronisation module	time synchronisation module; time synchronisation module manufacturer part number; time synchronisation module	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
					catalogue price GBP			
6	Defer Candidate-Only Sourcing	deferred	minor	account key switch	account key switch; account key switch manufacturer part number; account key switch catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	configuration backup module	configuration backup module; configuration backup module manufacturer part number; configuration backup module catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	firmware escrow tag	firmware escrow tag; firmware escrow tag manufacturer part number; firmware escrow tag catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	syslog collector	syslog collector; syslog collector manufacturer part number; syslog collector catalogue price GBP	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals
6	Defer Candidate-Only Sourcing	deferred	minor	VPN access appliance	VPN access appliance; VPN access appliance manufacturer part number; VPN access appliance	no immediate evidence required unless this candidate is promoted to procurement scope	no	candidate remains out of procurement totals

#	Batch	Status	Priority	Line	Search Starts	Required Evidence	Blocks Procurement	Target Outcome
					catalogue			
					price GBP			

Procurement Readiness Gate

procurement_not_started

0/7 procurement areas pass; 2 review, 5 blocked. Critical production-eligible BoM rows 0/6; procurement-blocking closure rows 6; active sourcing rows 6; deferred sourcing rows 198. Source quality no_sourcing_evidence; evidence authenticity no_evidence; costing costing_not_started; BoM admission candidate_bom_authorized. Can use for procurement: no. Next action: Collect production-quality source evidence for critical BoM rows before procurement use..

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
architecture review	review	0.33	architecture_freeze_review_required; structurally ready yes, independent review accepted no.	none	Collect accepted engineering/interface/requirement review evidence before procurement use.
bom evidence trace	blocked	0	0/6 critical rows production-eligible; 6 critical unsourced, 0 protocol-only, 0 source-reference blocked.	energy_storage_source_cell_string_lfp_prismatic_cells: Collect admissible source evidence for this critical line before BoM can pass. energy_storage_source_dc_string_output_dc_contactor_or_breaker: Collect admissible source evidence for this critical line before BoM can pass. energy_conversion_transduction_pcs_inverter_pcs_inverter: Collect admissible source evidence for this critical line before BoM can pass. control_compute_communication_bms_master_bms_master_controller: Collect admissible source evidence for this critical line before BoM can pass. safety_protection_fire_suppression_fire_detection_suppression: Collect admissible source evidence for this critical line before BoM can pass. plus 1 more	Collect production-quality source evidence for critical BoM rows before procurement use.
sourcing batch closure	blocked	0.9706	6 procurement-blocking closure row(s); 6 active sourcing row(s), 198 deferred row(s).	bom-close:control_compute_communication_bms_master_bms_master_controller: Collect admissible source evidence for this critical line before BoM can pass. bom-close:energy_conversion_transduction_pcs_inverter_pcs_inverter: Collect admissible source	Work the active sourcing batches until procurement-blocking closure rows are closed.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				evidence for this critical line before BoM can pass. bom-close:energy_storage_source_cell_string_lfp_prismatic_cells: Collect admissible source evidence for this critical line before BoM can pass. bom-close:energy_storage_source_dc_string_output_dc_contactor_or_breaker: Collect admissible source evidence for this critical line before BoM can pass. bom-close:mass_fluid_transport_process_pump_station_thermal_management_loop: Collect admissible source evidence for this critical line before BoM can pass. plus 1 more	
source reference quality	blocked	0	no_sourcing_evidence; 0/0 source-reference row(s) pass; 0 protocol, 0 placeholder.	none	Admit production source evidence before procurement readiness can start.
evidence authenticity	blocked	0	no_evidence; 0/0 evidence rows production-ready; 0 protocol.	No evidence records are present.	Add source and reviewer evidence before procurement readiness can start.
bom costing	blocked	0.14	costing_not_started ; critical priced 0/6; BoM total 0 GBP.	energy_storage_source_cell_string_lfp_prismatic_cells: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_storage_source_dc_string_output_dc_contactor_or_breaker: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass. energy_conversion_transduction_pcs_inverter_pcs_inverter: Collect admissible supplier/manufacturer/MPN/unit-cost evidence before BoM can pass.	Close costing blockers before procurement use.

Area	Verdict	Pass Ratio	Signal	Blockers	Required Action
				er/MPN/unit-cost evidence before BoM can pass. control_compute_c ommunication_bms _master_bms_mast er_controller: Collect admissible supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. safety_protection_fi re_suppression_fire _detection_suppres sion: Collect admissible supplier/manufactur er/MPN/unit-cost evidence before BoM can pass. plus 34 more	
prototype procurement policy	review	0	candidate_bom_aut horized; display mode candidate_only; prototype procurement use allowed no.	none	Prototype policy keeps procurement use disabled until external procurement review and approval are explicitly added.

Sourcing Evidence Pack

Evidence packets request source data only. They do not admit supplier, manufacturer, MPN or cost claims.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
LFP prismatic cells	1 candidate	LFP prismatic cells; LFP prismatic cells manufacturer part number; LFP prismatic cells catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
DC contactor or breaker	1 candidate	DC contactor or breaker; DC contactor or breaker manufacturer part number; DC contactor or breaker catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
PCS inverter	1 candidate	PCS inverter; PCS inverter manufacturer part number; PCS inverter catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
BMS master controller	1 candidate	BMS master controller; BMS master controller manufacturer part number; BMS master controller catalogue price GBP	componentWordId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWordId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Critical Line	Qty	Search Starts	Required Fields	Acceptance Criteria
Fire detection/suppression	1 candidate	Fire detection/suppression; Fire detection/suppression manufacturer part number; Fire detection/suppression catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.
Thermal management loop	1 candidate	Thermal management loop; Thermal management loop manufacturer part number; Thermal management loop catalogue price GBP	componentWorldId; supplierName; manufacturer; mpn; unitCostGbp; leadTimeWeeks; sourceGrade; evidence.ref; evidence.quote; retrievedAt	componentWorldId exactly matches a candidate BoM line. unitCostGbp is a positive GBP number with quantity basis clear enough to reproduce total cost. evidence.kind is source, with a retrievable URL, quote reference, catalogue page, distributor listing, or quote file reference.

Cost model status: pending sourcing. Architecture is being reviewed before cost claims are admitted.

Cost Bucket	Estimate
CAPEX	£0
Annual OPEX	£0
NRE	£25,000

No admitted supplier spend yet. Supplier rows appear only after source-backed evidence records are admitted.

Regulatory And Risk

Standard	Title	Jurisdiction	Evidence Required
IEC-62619	Industrial lithium battery safety	Global / UK-adopted	Cell certificates and pack-level safety validation.
UL-9540A	Thermal runaway fire propagation test	US / investor diligence	Cell, module and unit-level propagation evidence.
G99	UK grid connection requirements	UK	Protection settings, relay test certificates and commissioning records.

Hazard	S	O	D	RPN	Mitigation
Thermal runaway propagation	10	3	4	120	Cell spacing, thermal monitoring and clean-agent suppression.
DC arc fault	9	3	4	108	Arc detection, contactor isolation and maintenance access controls.

Section Issues And Score

bom

- blocker / critical_part_unpriced: LFP prismatic cells is critical but has no credible unit cost.
- blocker / critical_part_unpriced: DC contactor or breaker is critical but has no credible unit cost.
- blocker / critical_part_unpriced: PCS inverter is critical but has no credible unit cost.
- blocker / critical_part_unpriced: BMS master controller is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Fire detection/suppression is critical but has no credible unit cost.
- blocker / critical_part_unpriced: Thermal management loop is critical but has no credible unit cost.
- major / low_priced_line_ratio: Only 0% of BoM lines have prices.

Section Score Ledger

Rule: start at 9, subtract 3 per blocker, 1.5 per major, 0.5 per minor, floor at 1. Mean score 7.67. This is a deterministic gate score from current validators, not an external reviewer score or proof of real-world engineering quality. Reviewer acceptance is tracked separately in the verification ledger.

Section	Base	B/M/m Issues	Deduction	Final	Target	Issue Deductions	Rationale
executive_summary	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
brief_requirements	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
design_modules	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
bom	9	6/1/0	-19.5	1	below target	blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); blocker/critical_part_unpriced (-3); major/low_priced_line_ratio (-1.5)	Score is base minus deterministic deductions for attached section issues.
feasibility_notes	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
sources_references	9	0/0/0	-0	9	pass	none	No current deterministic validator issue is attached to this scored section.
Mean	7.67						